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FACULTY OF ENGINEERING

DEPARTMENT OF ELECTRICAL AND INFORMATION ENGINEERING

Addressing Heatwaves, Air Pollution, and Human Health in West and East Africa: High-Frequency Sensor Monitoring and AI-Driven Analysis

MSc Proposal

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FEE6/49465/2025

[BSc. Electrical and Electronics Engineering (UoN)]

A Research Proposal Submitted in Partial Fulfilment of the
Requirements for the Award of Master of Science Degree in
Nuclear Science and Technology at the Department of Electrical
and Information Engineering of the University of Nairobi.

March, 2026

APPROVALS

Supervisors' Approval

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ABBREVIATIONS AND ACRONYMS

AI	Artificial Intelligence
ANN	Artificial Neural Network
API	Application Programming Interface
AQI	Air Quality Index
BiLSTM	Bidirectional Long Short-Term Memory
CNN	Convolutional Neural Network
CO	Carbon Monoxide
CO ₂	Carbon Dioxide
DL	Deep Learning
DPSIR	Driver-Pressure-State-Impact-Response
DT	Decision Tree
GANs	Generative Adversarial Networks
GIS	Geographic Information System
GPR	Gaussian Process Regression
GRU	Gated Recurrent Unit
HVI	Heat Vulnerability Index
IoT	Internet of Things
LoRaWAN	Long Range Wide Area Network
LSTM	Long Short-Term Memory
LULC	Land Use and Land Cover
MAE	Mean Absolute Error
ML	Machine Learning
MQTT	Message Queuing Telemetry Transport
NO ₂	Nitrogen Dioxide
O ₃	Ozone
PM	Particulate Matter
PM ₁₀	Particulate Matter $\leq 10 \mu\text{m}$
PM _{2.5}	Particulate Matter $\leq 2.5 \mu\text{m}$
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
R ²	Coefficient of Determination (R-squared)
RF	Random Forest
RMSE	Root Mean Square Error
SDG	Sustainable Development Goal
SHAP	SHapley Additive exPlanations
SVM	Support Vector Machine
SVR	Support Vector Regression
UHI	Urban Heat Island
XAI	Explainable Artificial Intelligence
XGBoost	Extreme Gradient Boosting

ABSTRACT

Rapid urbanization in Africa has intensified extreme heat events and air pollution, posing serious health risks to vulnerable populations. This proposal outlines a cloud-based, AI-enhanced monitoring platform that integrates real-time environmental data (PM_{2.5}, PM₁₀, CO₂, NO₂, O₃, temperature, humidity, wind) with health records and socio-economic indicators. Using Llama2 for anomaly detection and forecasting, the system will generate timely alerts, vulnerability maps, and policy briefs. Key objectives include deploying sensor networks in target cities (Accra, Ghana and Nairobi, Kenya), correlating pollution and heat with hospital data, building local technical capacity, and producing replicable heat action plans. The initiative aims to improve health resilience, guide adaptive urban planning, and foster digital transformation in West and East Africa.

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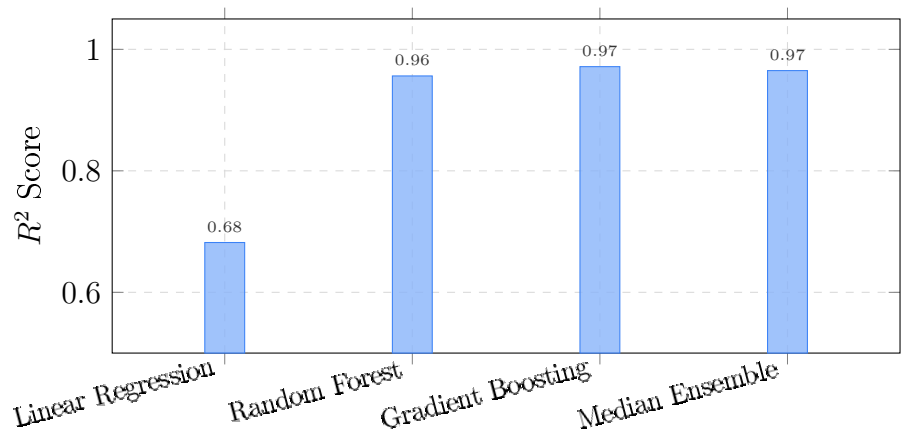


Figure 1: Comparative R^2 scores across the four model approaches. Gradient Boosting achieves the highest individual score (0.9715), while the median ensemble provides variance-reduced production stability.

Table 1: Structured Literature Review: 20 Most Relevant Studies on Heatwaves, Air Pollution, and AI in Africa

Author(s) & Year	Title / Focus	Key Methodology	Key Findings	Limitations & Gaps	Relevance to Our Proposal
Bhaskar & Valli Mayil (2025) [?]	Traffic air pollution forecasting using ML	Review of ML/DL/hybrid models for PM _{2.5} , NO ₂ , CO	Traffic is major pollution source; ML provides actionable insights	No integration with health or heat variables	Informs traffic–pollution correlation and sensor placement
Gaur & Deb (2026) [?]	ML for UHI assessment: comprehensive review	Systematic review of 93 studies; RMSE, MAE, R ² , SHAP	Tropical cities only 9% of studies; <8% use real-time IoT; 5% evaluate equity	Equity gap; tropical Africa underrepresented	Confirms geographic and equity gaps; justifies HVI mapping
Bainomugisha et al. (2024) [?]	AirQo: AI-driven sensor networks in Africa	Custom low-cost PM monitors; 10+ cities in 8 African countries	National air quality standards; city action plans	PM only; no NO ₂ , O ₃ , heat stress, or health data integration	Closest African precedent; extends AirQo to multi-pollutant + health
Kalisa & Sudmant (2025) [?]	Heatwaves amplify air pollution in SSA	Low-cost sensors in Kigali; PM _{2.5} , NO ₂ , O ₃ analysis	O ₃ peaks 40% higher during heatwaves; PM _{2.5} driven by local emissions	Single city; no health data integration	First evidence of heat-pollution synergy in SSA; justifies multi-pollutant monitoring

Table 1 – continued

Author(s) & Year	Title / Focus	Key Methodology	Key Findings	Limitations & Gaps	Relevance to Our Proposal
Touré et al. (2025) [?]	ML prediction of heatwave hospitalizations in Senegal	RF, XGBoost, GAM; hospital data 2017–2022	RF $R^2=0.51$ – 0.72 ; 3–5 day lag to hospitalization peak	Single hospital; no air pollution covariates	Demonstrates ML for heat-health prediction in West Africa; supports health data integration
Adewumi & Ajayi (2025) [?]	Climate-informed ML for heat-related health risks in Nigeria	Transformer, LSTM; 52,846 records from 2,300 patients	Transformer accuracy 92.7%; potential 28–32% admission reduction	Limited to one region; requires personal health data	Validates transformer models; informs Llama2 fine-tuning approach
Ramsay et al. (2024) [?]	Humid heat stress overlooked in informal settlements	In situ monitoring across 7 cities; wet-bulb temperature analysis	Informal settlements experience higher Tw than weather stations	Lack of local climate data; no health integration	Highlights exposure inequity; justifies HVI mapping and targeting informal settlements
Awokola et al. (2022) [?]	Longitudinal $PM_{2.5}$ in 8 SSA countries using low-cost sensors	PurpleAir-II-SD network; 15 sites, 11 cities; harmonic regression	Annual $PM_{2.5}$ 10–116 $\mu g/m^3$; diurnal traffic peaks	Low-cost sensor accuracy; limited to $PM_{2.5}$	Establishes $PM_{2.5}$ baseline; informs sensor deployment strategy

Table 1 – continued

Author(s) & Year	Title / Focus	Key Methodology	Key Findings	Limitations & Gaps	Relevance to Our Proposal
Van de Walle et al. (2021) [?]]	Vegetation and heat exposure in Kampala dense settlements	Low-cost sensor network; Humidex index; NDVI from satellite	Daily max Humidex difference up to 14.5°C; NDVI explains 75% variation	Single city; limited time period	Quantifies intra-urban heat inequality; supports urban greening recommendations
Dey, Arora & Tripathi (2022) [?]]	Low-cost sensor calibration via semi-supervised domain adaptation	HL+WMME; 11 Mumbai PM _{2.5} locations	Avg. R ² =69.5% vs 48% baseline; leverages unlabelled data	Validated only on PM _{2.5} in Mumbai; sensor drift not addressed	Enables calibration with minimal labelled data in African deployments
Garcia et al. (2025) [?]]	IoT-based air quality and AI: systematic review	PRISMA review of 147 studies; 5 AI application areas	DL (LSTM, transformers) outperform traditional methods; edge computing gap	Sensor drift; DL edge deployment; limited policy translation	Validated AI taxonomy; edge computing gap motivates edge-cloud hybrid design
Bekkar et al. (2021) [?]]	Air-pollution prediction with deep learning	CNN-LSTM; Beijing PM _{2.5} ; 420,768 instances	CNN-LSTM: MAE=6.742, RMSE=12.921, R ² =0.989	Single city; high-quality data required; no generalizability analysis	CNN-LSTM architecture adopted for PM _{2.5} and temperature forecasting

Table 1 – continued

Author(s) & Year	Title / Focus	Key Methodology	Key Findings	Limitations & Gaps	Relevance to Our Proposal
Assaf, Hu & Assaad (2023) [?]	Predicting UHI severity using Bayesian networks	White-box Bayesian network; demographic, meteorological, LULC factors	87.88% accuracy predicting UHI severity at census-tract level	Limited to New Jersey; relies on expert knowledge; census-tract resolution	Informs vulnerability mapping; LULC factors relevant to urban heat mitigation
Ikegwu et al. (2026) [?]	AI expert systems for environmental surveillance	PRISMA review; 26 primary studies	Hybrid AI enables real-time alerts; edge AI critical for resource-constrained settings	Poor integration with national frameworks; Africa underrepresented	Validates hybrid AI + rule-based approach; governance gap addressed by heat action plans
Mehryar, Yazdanpanah & Tong (2024) [?]	AI and climate resilience governance	Systematic review (244 articles) + expert survey	64% focus on risk assessment; only 6% on implementation	Insufficient socio-economic data; interpretability; capacity gaps	Frames AI for climate governance; implementation gap motivates policy brief deliverables
Ramadan et al. (2024) [?]	Real-time IoT-AI for industrial air pollution	15-sensor node; LSTM, RF; 3 months, 5-min intervals	LSTM $R^2=0.99$ (temp); RF $R^2=0.84$ ($PM_{2.5}$); proactive fan control	Single industrial site; no long-term drift analysis	Validates real-time IoT+AI with proactive alerts; informs forecasting architecture

Table 1 – continued

Author(s) & Year	Title / Focus	Key Methodology	Key Findings	Limitations & Gaps	Relevance to Our Proposal
Ariff, Bakar & Lim (2023) [?]	PM ₁₀ prediction using K-means + LSTM hybrid	K-means clustering + multivariate LSTM; 60 stations	Training time reduced from 4,952 s to 168 s; 71.7% stations maintain accuracy	Meteorological features not included	Efficient scalable forecasting across many nodes; clustering strategy adapted for multi-city network
He et al. (2024) [?]	Lightweight IoT intrusion detection	Feature grouping; DT, RF, KNN, XGB; 3 IoT datasets	>99.5% accuracy; low parsing/training time vs DL	Tested on botnet/DoS attacks; extensibility to new protocols needs validation	Lightweight IDS protects sensor data streams; complements Llama2 anomaly detection
Yasin et al. (2022) [?]	IoT IAQ monitoring using off-the-shelf devices	COTS sensors; Node-RED, InfluxDB, Grafana; 4-site validation	Generic replicable IoT architecture; unified data ingest	Sensor accuracy variability; no standardized calibration	Practical low-cost IoT stack; InfluxDB/Grafana architecture directly adopted
Tetali (2025) [?]	AI in heat mitigation and adaptation	Review of 65 studies; DPSIR framework	AI used for heatwave/UHI prediction (46%); gap in tropical/Global South	Limited research in tropical regions; lack of practical implementation	Directly supports Africa UHI focus; DPSIR framework adopted for policy briefs

CHAPTER 1

INTRODUCTION

1.1 Background of Study

Rapid urbanization across Africa has led to a significant increase in both extreme heat events and localized air pollution. In metropolitan areas, vehicular traffic is a primary contributor to poor air quality, emitting particulate matter (PM_{2.5}, PM₁₀), nitrogen oxides (NO₂), and carbon monoxide (CO) [?]. These environmental stressors often co-occur, creating hazardous conditions for urban residents, particularly in vulnerable socio-economic brackets.

The convergence of rapid population growth, informal urban expansion, and intensifying climate change has rendered many African cities particularly susceptible to compound environmental risks. The Urban Heat Island (UHI) effect—whereby impervious surfaces and reduced vegetation cover cause urban areas to retain significantly more heat than surrounding rural zones—is exacerbated in cities with high traffic density and inadequate green infrastructure [? ?]. These conditions disproportionately affect low-income populations with limited access to cooling and healthcare, making environmental monitoring and early warning systems a critical public health priority.

Globally, AI-driven environmental platforms have demonstrated their potential to integrate sensor data with health and socio-economic indicators for evidence-based policy [? ?]. However, tropical Africa remains severely underrepresented in such deployments, with existing platforms largely constrained to particulate matter monitoring and lacking integration with heat stress indices and hospital outcome data [? ?]. This proposal directly addresses these gaps.

1.2 Problem Statement

Despite the growing risks, many African cities lack robust, high-frequency environmental monitoring networks. Current systems often rely on sparse, unreliable sensors with high maintenance costs and limited integration with health or meteorological data. There is a critical gap in understanding the causal links between specific emission sources (e.g., vehicular traffic), heatwave severity, and hospital admissions for heat-related illnesses. Without real-time, actionable intelligence, policy makers and city planners cannot implement effective adaptive measures.

Gaur and Deb [?] confirm that tropical cities account for only 9 % of UHI mitigation studies and fewer than 8 % use real-time IoT sensors, while Ikegwu et al. [?] note that tropical Africa is severely underrepresented in AI expert-system deployments for environmental surveillance. Furthermore, most deployed systems—including the well-established AirQo platform [?]—focus solely on PM and do not integrate multi-pollutant monitoring with health outcomes and socio-economic vulnerability data. This proposal directly addresses these gaps.

1.3 Objectives

1.3.1 Overall Objective

The overall objective of this research is to develop a cloud-based, AI-enhanced monitoring platform that integrates high-frequency environmental data with health and socio-economic indicators to improve climate resilience in West and East African cities.

1.3.2 Specific Objectives

- i) To deploy high-frequency sensor networks (PM_{2.5}, PM₁₀, CO₂, NO₂, O₃, temperature, humidity, wind) at 5-minute intervals in target urban areas.
- ii) To correlate real-time environmental data with hospital records on heat-related illnesses (e.g., heatstroke) and socio-economic vulnerability profiles.
- iii) To leverage AI (Llama2) for real-time anomaly detection, hazard forecasting, and automated alerting.
- iv) To formulate evidence-based policy recommendations, including heat action plans and urban planning adjustments (e.g., low-emission zones, congestion pricing).
- v) To strengthen local technical capacity in IoT deployment, big data management, and AI-driven climate resilience.

1.4 Research Questions

1. How effectively can a low-cost IoT sensor network provide reliable, high-frequency air quality and heat stress data in resource-constrained West and East African urban environments?

2. What is the quantitative correlation between real-time PM_{2.5}/temperature levels and hospitalizations for respiratory and heat-related distress?
3. Can transformer-based models (e.g., Llama2) reliably detect anomalies and forecast environmental hazards using low-cost sensor data?

1.5 Justification and Significance of the Research

Justification. This study addresses the urgent need for local, data-driven climate adaptation. By integrating low-cost sensor technology with advanced AI, the research provides a scalable solution for cities with limited resources. The established AirQo platform in East Africa demonstrates the feasibility of such deployments—having monitored air quality in more than 10 cities across 8 African countries [?]—yet it does not integrate heat stress, health records, or socio-economic vulnerability into a unified predictive framework. This proposal explicitly fills that gap.

Significance. The research will benefit city officials, health providers, and the general public by providing early warnings and vulnerability maps. It contributes to the digital transformation of African environmental governance and offers a replicable model for urban resilience. The outcomes directly support the United Nations Sustainable Development Goals (SDGs): SDG 3 (Good Health and Well-being), SDG 11 (Sustainable Cities and Communities), and SDG 13 (Climate Action).

1.6 Scope of the Work

The research focuses on selected major cities in West Africa (Accra, Ghana) and East Africa (Nairobi, Kenya) during the period 2026–2027. It includes the deployment of IoT sensor networks, development of a cloud-based AI platform, and analysis of integrated health and socio-economic datasets. The study is limited to urban environments and does not extend to rural or peri-urban areas. Health data analysis is restricted to publicly accessible, anonymized hospital records pertaining to heat-related and respiratory admissions.

1.7 Organization of the Proposal

This proposal is organized into five chapters. Chapter 1 introduces the background, problem statement, objectives, and research questions. Chapter 2 reviews relevant literature on ML in smart cities, air quality monitoring, and UHI assessment. Chapter 3 describes the research methodology, sensor specifications, cloud architecture, and AI

pipeline. Chapter 4 outlines the expected outcomes. Chapter 5 presents the work plan and budget.

CHAPTER 2

LITERATURE REVIEW

2.1 Overview of the Chapter

This chapter reviews recent advances in machine learning for urban air quality monitoring and heat island assessment, with particular emphasis on the challenges of data reliability, urban complexity, and knowledge gaps in the African context. The review is structured around four thematic areas: (i) ML in air quality forecasting, (ii) smart city AI trade-offs, (iii) UHI prediction, and (iv) IoT-based monitoring platforms. A structured synthesis table (Table 2.1) provides a comparative overview of the **20 most relevant** studies to this research. Additional relevant studies are cited throughout the narrative to provide comprehensive coverage of the literature. The chapter concludes with a summary of the identified research gaps that this proposal addresses.

2.2 Machine Learning in Air Quality Forecasting

Vehicular traffic is one of the dominant sources of urban air pollution, emitting $\text{PM}_{2.5}$, NO_2 , and CO at levels that frequently exceed safe exposure thresholds in metropolitan areas. Bhaskar and Valli Mayil [?] synthesized ML methods—including deep learning and hybrid models—applied to traffic-induced pollution forecasting. Their review confirms the predictive value of ML for policy intervention but highlights persistent challenges: sensor unreliability, high computational costs, and a critical absence of integration between traffic data, health records, and thermal variables.

Essamlali, Nhailla and El Khaili [?] conducted a PRISMA-compliant systematic review of supervised learning approaches for predicting key pollutants. Ensemble methods—particularly Random Forest (RF) and XGBoost—alongside LSTM networks dominate the literature due to high accuracy, while hybrid models offer further improvements in forecasting. Data scarcity, sensor unreliability, and limited generalizability across geographic regions remain major barriers.

Ariff, Bakar and Lim [?] demonstrated a hybrid K-means clustering plus LSTM model for PM_{10} forecasting across 60 Malaysian monitoring stations, reducing total training time from 4,952 s to 168 s while maintaining prediction accuracy. This efficiency gain is critical for resource-constrained deployments.

Garcia et al. [?] conducted a PRISMA review of 147 IoT-based air quality and AI studies, categorizing AI applications into five areas: data imputation, sensor calibration, anomaly detection, AQI estimation, and short-term forecasting. Deep learning models outperform traditional methods, and hybrid architectures achieve the highest accuracy. Edge computing was identified as an emerging gap—directly motivating the edge-cloud architecture proposed here.

Bekkar et al. [?] validated a CNN-LSTM architecture on four years of Beijing $\text{PM}_{2.5}$ data, demonstrating superiority over individual LSTM and CNN approaches ($\text{MAE} = 6.742$, $\text{RMSE} = 12.921$, $R^2 = 0.989$). Spatial-temporal feature extraction across adjacent stations improved accuracy substantially—a key insight informing the multi-node sensor network design.

Dey, Arora and Tripathi [?] addressed calibrating low-cost sensors against sparse reference monitors using semi-supervised domain adaptation. Their proposed HL+WMME method achieves an average R^2 of 69.5% versus 48% for the next-best baseline, leveraging abundant unlabelled sensor data. This approach is particularly relevant where co-located reference monitors are available only during a brief calibration window.

Kalisa and Sudmant [?] provided the first direct evidence of heat-pollution synergy in Sub-Saharan Africa. Using low-cost sensors in Kigali, Rwanda, they found O_3 peaks up to 40% higher during heatwaves, while $\text{PM}_{2.5}$ and NO_2 were driven more by local emissions. This study justifies the multi-pollutant monitoring approach and the need for heat-pollution co-analysis.

Awokola et al. [?] established a baseline for $\text{PM}_{2.5}$ across 11 cities in 8 SSA countries using PurpleAir-II-SD sensors. Annual concentrations ranged from 10 to $116 \mu\text{g}/\text{m}^3$, with diurnal peaks around rush hour, confirming traffic as a primary emission source and informing sensor deployment strategy.

Additional studies on air quality in African cities include Ninsiima et al. [?] who reported consistently high $\text{PM}_{2.5}$ levels in Kampala (average $59 \mu\text{g}/\text{m}^3$), Saetae et al. [?] who identified seasonal $\text{PM}_{2.5}$ hotspots in Akure, Nigeria, and Doe et al. [?] who mapped health risks in Accra using geospatial regression. Odirichukwu et al. [?] reviewed the potential of AI-driven IoT for air quality management in Nigeria, while Tshuma et al. [?] demonstrated a cost-effective AI air quality system in South Africa.

2.3 Smart Cities and AI Trade-offs

Band et al. [?] conducted an in-depth systematic review of ML in smart city applications. While DL and ensemble methods achieve the highest accuracy for complex environmental data, they impose significant computational overhead—a critical trade-off for real-time monitoring in African cities with limited infrastructure. This work guides the model selection strategy, justifying the use of a parameter-efficient Llama2-7B (LoRA adapter).

Kaied, Alhosani and Darwish [?] demonstrated an integrated IoT-AI framework for urban air quality management in Ajman, UAE—a hot-arid environment analogous to parts of West Africa. Their system successfully bridged technical prediction capabilities with actionable policy outputs.

Sarker and Jahan [?] reviewed 142 studies on AI-driven MIS applications for environmental risk monitoring, confirming that more than 60 % of AI models achieve above 85 % prediction accuracy when integrated with GIS and MIS frameworks. MIS platforms serve as operational backbones for data integration and multi-agency coordination.

Mehryar, Yazdanpanah and Tong [?] analysed AI for climate resilience governance, finding that 64 % of studies focus on risk assessment while only 6 % address implementation. Key challenges include insufficient socio-economic data, model interpretability, and organizational capacity—motivating the proposed integration of socio-economic vulnerability indicators and human-in-the-loop alert design.

Eze et al. [?] reviewed AI for climate adaptation in Africa, noting successful deployments in Kenya and Ethiopia but highlighting data scarcity and AI energy footprint as concerns. Miller et al. [?] examined AI agents with IoT for environmental monitoring, emphasizing interoperability and security challenges. Liu et al. [?] explored deep transfer learning for industrial IoT, enabling data-efficient model retraining relevant to African contexts.

2.4 Urban Heat Island Prediction

Assaf, Hu and Assaad [?] demonstrated interpretable white-box Bayesian networks for predicting UHI severity at the census-tract level (87.88 % accuracy), integrating demographic, meteorological, and LULC factors. Their methodology informs the spatial vulnerability mapping component of this proposal.

Gaur and Deb [?] conducted a systematic review of 93 UHI studies, finding that RF and neural networks are most frequently applied, while hybrid neural-physical

models reduce error by 23–41 %. Crucially, tropical cities account for only 9 % of mitigation studies, fewer than 8 % use real-time IoT sensors, and only 5 % evaluate equity—directly positioning this proposal in an underserved research space.

Tetali [?] reviewed 65 studies on AI in heat mitigation and adaptation, using the DPSIR framework. The study explicitly calls for greater research in tropical regions and the Global South—the precise focus of this proposal.

Hashir et al. [?] reviewed AI for indoor air quality in educational settings, confirming RF and XGBoost as optimal for low-cost sensor deployment. Afolabi et al. [?] demonstrated the integration of remote sensing, GIS, and AI to model UHI in Lagos, showing LST increases of 3.2°C to 5.4°C over a decade and validating Gradient Boosted Trees for LST prediction ($R^2=0.95$).

Ramsay et al. [?] highlighted that humid heat stress is significantly underestimated in urban informal settlements, where residents experience higher wet-bulb temperatures than those recorded at nearby meteorological stations. This study underscores the need for HVI mapping and targeted interventions in informal settlements.

Van de Walle et al. [?] quantified intra-urban heat inequality in Kampala, finding a daily max Humidex difference of up to 14.5°C between stations, with NDVI explaining 75 % of variation. This supports urban greening recommendations in policy briefs. Sampson et al. [?] similarly used low-cost sensors and citizen science to assess thermal inequality in African slums.

Bello Hafisat Omodasola [?] proposed a GeoAI-enabled framework for monitoring urban climate shocks in five African cities, emphasizing the need for ethical principles and participatory mapping. Jack et al. [?] described the HE²AT Center protocol for leveraging data science and machine learning for heat-health adaptation in Johannesburg and Abidjan.

2.5 IoT-Based Monitoring Platforms

Bainomugisha et al. [?] documented AirQo, an AI-driven IoT monitoring system deployed across more than 10 cities in 8 African countries. This is the most directly relevant precedent for the proposed platform; AirQo covers only PM and does not integrate NO₂, O₃, heat stress, or hospital data—gaps explicitly addressed here.

Ramadan et al. [?] validated a real-time IoT-AI system at an industrial site in Istanbul using LSTM and RF for multi-pollutant forecasting. The system proactively activates exhaust fans upon high-pollution predictions, demonstrating a direct feedback loop between sensing and intervention.

García et al. [?] deployed a solar-powered, LoRaWAN-based IoT system in an industrial environment. The off-grid, tamper-resistant design is directly relevant to the proposed African sensor nodes.

Yasin et al. [?] demonstrated a replicable IoT architecture for IAQ monitoring using commercial off-the-shelf sensors, unified through Node-RED, InfluxDB, and Grafana. This stack is directly adopted in the proposed cloud platform.

He et al. [?] proposed a lightweight feature-grouped intrusion detection system achieving $>99.5\%$ accuracy and F1-score, offering a computationally efficient mechanism for protecting IoT sensor data streams in resource-constrained deployments.

Gueye et al. [?] deployed an IoT and mapping-based approach to identify $\text{PM}_{2.5}$ hotspots in Dakar, demonstrating the feasibility of mobile monitoring with GPS-equipped vehicles. Kumar and Bharti [?] presented a fine-scale air pollution exposure mapping framework using low-cost sensor networks, while Johnson et al. [?] provided additional longitudinal $\text{PM}_{2.5}$ data from the PurpleAir network.

2.6 Health Impact and Predictive Modeling

Touré et al. [?] demonstrated the feasibility of ML for heat-health prediction in West Africa. Using RF, XGBoost, and GAM on hospital admissions in Matam, Senegal, they found a 3–5 day lag between heatwave events and hospitalization peaks, with RF achieving $R^2 = 0.51\text{--}0.72$. This supports health data integration in the proposed platform.

Adewumi and Ajayi [?] validated transformer models for personalized heat-health risk prediction in Nigeria. Their Transformer model achieved 92.7% accuracy, with potential to reduce admissions by $28\text{--}32\%$. This informs the Llama2 fine-tuning approach.

Baharane and Shatalov [?] assessed health impacts of air pollution in East African countries, finding a 14.26% decrease in attributable deaths over 30 years but highlighting household air pollution as a persistent concern. Bililign et al. [?] outlined a framework for air quality measurement and modeling in East African megacities, emphasizing the need for capacity building and partnerships.

2.7 Summary of Literature and Research Gap

The reviewed literature confirms that AI models are powerful tools for environmental monitoring but reveals a distinct set of gaps that the proposed research addresses:

1. **Geographic gap.** Tropical Africa is severely underrepresented in AI-driven UHI and air quality studies, with tropical cities comprising only 9 % of UHI mitigation studies [?].
2. **Integration gap.** No existing African platform integrates multi-pollutant monitoring ($\text{PM}_{2.5}$, PM_{10} , NO_2 , O_3 , CO_2), heat stress metrics, hospital health data, and socio-economic vulnerability into a single predictive framework [?].
3. **Validation gap.** Few studies provide long-term operational validation in resource-constrained settings, and most AI models are tested only on historical datasets or short pilots [?].
4. **Equity gap.** Only 5 % of reviewed studies evaluate equity dimensions of environmental exposure [?].
5. **Governance gap.** Only 6 % of AI-climate studies address implementation, leaving the translation of predictive insights into actionable policy largely unaddressed [?].

This research directly targets all five gaps by deploying a real-time, multi-pollutant, health-integrated monitoring platform with embedded equity analysis and structured policy output mechanisms in West and East African cities.

Table 2.1: Structured Literature Review: 20 Most Relevant Studies on Heatwaves, Air Pollution, and AI in Africa

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Bhaskar & Valli Mayil (2025) [?]	Traffic air pollution forecasting using ML	Review of ML/DL/hybrid models for PM _{2.5} , NO ₂ , CO	Traffic is major pollution source; ML provides actionable insights	No integration with health or heat variables	Informs traffic–pollution correlation and sensor placement
Gaur & Deb (2026) [?]	ML for UHI assessment: comprehensive review	Systematic review of 93 studies; RMSE, MAE, R ² , SHAP	Tropical cities only 9% of studies; <8% use real-time IoT; 5% evaluate equity	Equity gap; tropical Africa underrepresented	Confirms geographic and equity gaps; justifies HVI mapping
Bainomugisha et al. (2024) [?]	AirQo: AI-driven sensor networks in Africa	Custom low-cost PM monitors; 10+ cities in 8 African countries	National air quality standards; city action plans	PM only; no NO ₂ , O ₃ , heat stress, or health data integration	Closest African precedent; extends AirQo to multi-pollutant + health
Kalisa & Sudmant (2025) [?]	Heatwaves amplify air pollution in SSA	Low-cost sensors in Kigali; PM _{2.5} , NO ₂ , O ₃ analysis	O ₃ peaks 40% higher during heatwaves; PM _{2.5} driven by local emissions	Single city; no health data integration	First evidence of heat-pollution synergy in SSA; justifies multi-pollutant monitoring

Table 2.1 – continued

Author(s) & Year	Title / Focus	Key Methodology	Key Findings	Limitations & Gaps	Relevance to Our Proposal
Touré et al. (2025) [?]	ML prediction of heatwave hospitalizations in Senegal	RF, XGBoost, GAM; hospital data 2017–2022	RF $R^2=0.51$ – 0.72 ; 3–5 day lag to hospitalization peak	Single hospital; no air pollution covariates	Demonstrates ML for heat-health prediction in West Africa; supports health data integration
Adewumi & Ajayi (2025) [?]	Climate-informed ML for heat-related health risks in Nigeria	Transformer, LSTM; 52,846 records from 2,300 patients	Transformer accuracy 92.7%; potential 28–32% admission reduction	Limited to one region; requires personal health data	Validates transformer models; informs Llama2 fine-tuning approach
Ramsay et al. (2024) [?]	Humid heat stress overlooked in informal settlements	In situ monitoring across 7 cities; wet-bulb temperature analysis	Informal settlements experience higher T_w than weather stations	Lack of local climate data; no health integration	Highlights exposure inequity; justifies HVI mapping and targeting informal settlements
Awokola et al. (2022) [?]	Longitudinal $PM_{2.5}$ in 8 SSA countries using low-cost sensors	PurpleAir-II-SD network; 15 sites, 11 cities; harmonic regression	Annual $PM_{2.5}$ 10–116 $\mu g/m^3$; diurnal traffic peaks	Low-cost sensor accuracy; limited to $PM_{2.5}$	Establishes $PM_{2.5}$ baseline; informs sensor deployment strategy

Table 2.1 – continued

Author(s) & Year	Title / Focus	Key Methodology	Key Findings	Limitations & Gaps	Relevance to Our Proposal
Van de Walle et al. (2021) [?]]	Vegetation and heat exposure in Kampala dense settlements	Low-cost sensor network; Humidex index; NDVI from satellite	Daily max Humidex difference up to 14.5°C; NDVI explains 75% variation	Single city; limited time period	Quantifies intra-urban heat inequality; supports urban greening recommendations
Dey, Arora & Tripathi (2022) [?]]	Low-cost sensor calibration via semi-supervised domain adaptation	HL+WMME; 11 Mumbai PM _{2.5} locations	Avg. R ² =69.5% vs 48% baseline; leverages unlabelled data	Validated only on PM _{2.5} in Mumbai; sensor drift not addressed	Enables calibration with minimal labelled data in African deployments
Garcia et al. (2025) [?]]	IoT-based air quality and AI: systematic review	PRISMA review of 147 studies; 5 AI application areas	DL (LSTM, transformers) outperform traditional methods; edge computing gap	Sensor drift; DL edge deployment; limited policy translation	Validated AI taxonomy; edge computing gap motivates edge-cloud hybrid design
Bekkar et al. (2021) [?]]	Air-pollution prediction with deep learning	CNN-LSTM; Beijing PM _{2.5} ; 420,768 instances	CNN-LSTM: MAE=6.742, RMSE=12.921, R ² =0.989	Single city; high-quality data required; no generalizability analysis	CNN-LSTM architecture adopted for PM _{2.5} and temperature forecasting

Table 2.1 – continued

Author(s) & Year	Title / Focus	Key Methodology	Key Findings	Limitations & Gaps	Relevance to Our Proposal
Assaf, Hu & Assaad (2023) [?]	Predicting UHI severity using Bayesian networks	White-box Bayesian network; demographic, meteorological, LULC factors	87.88% accuracy predicting UHI severity at census-tract level	Limited to New Jersey; relies on expert knowledge; census-tract resolution	Informs vulnerability mapping; LULC factors relevant to urban heat mitigation
Ikegwu et al. (2026) [?]	AI expert systems for environmental surveillance	PRISMA review; 26 primary studies	Hybrid AI enables real-time alerts; edge AI critical for resource-constrained settings	Poor integration with national frameworks; Africa underrepresented	Validates hybrid AI + rule-based approach; governance gap addressed by heat action plans
Mehryar, Yazdanpanah & Tong (2024) [?]	AI and climate resilience governance	Systematic review (244 articles) + expert survey	64% focus on risk assessment; only 6% on implementation	Insufficient socio-economic data; interpretability; capacity gaps	Frames AI for climate governance; implementation gap motivates policy brief deliverables
Ramadan et al. (2024) [?]	Real-time IoT-AI for industrial air pollution	15-sensor node; LSTM, RF; 3 months, 5-min intervals	LSTM $R^2=0.99$ (temp); RF $R^2=0.84$ ($PM_{2.5}$); proactive fan control	Single industrial site; no long-term drift analysis	Validates real-time IoT+AI with proactive alerts; informs forecasting architecture

Table 2.1 – continued

Author(s) & Year	Title / Focus	Key Methodology	Key Findings	Limitations & Gaps	Relevance to Our Proposal
Ariff, Bakar & Lim (2023) [?]	PM ₁₀ prediction using K-means + LSTM hybrid	K-means clustering + multivariate LSTM; 60 stations	Training time reduced from 4,952 s to 168 s; 71.7% stations maintain accuracy	Meteorological features not included	Efficient scalable forecasting across many nodes; clustering strategy adapted for multi-city network
He et al. (2024) [?]	Lightweight IoT intrusion detection	Feature grouping; DT, RF, KNN, XGB; 3 IoT datasets	>99.5% accuracy; low parsing/training time vs DL	Tested on botnet/DoS attacks; extensibility to new protocols needs validation	Lightweight IDS protects sensor data streams; complements Llama2 anomaly detection
Yasin et al. (2022) [?]	IoT IAQ monitoring using off-the-shelf devices	COTS sensors; Node-RED, InfluxDB, Grafana; 4-site validation	Generic replicable IoT architecture; unified data ingest	Sensor accuracy variability; no standardized calibration	Practical low-cost IoT stack; InfluxDB/Grafana architecture directly adopted
Tetali (2025) [?]	AI in heat mitigation and adaptation	Review of 65 studies; DPSIR framework	AI used for heatwave/UHI prediction (46%); gap in tropical/Global South	Limited research in tropical regions; lack of practical implementation	Directly supports Africa UHI focus; DPSIR framework adopted for policy briefs

CHAPTER 3

METHODOLOGY

3.1 Overview of the Chapter

This chapter describes the research design, study area, sampling strategy, data collection instruments (IoT sensor network and cloud platform), the AI-driven analysis framework, and ethical safeguards.

3.2 Research Design

The study adopts a mixed-methods, quasi-experimental design combining continuous quantitative monitoring with qualitative stakeholder engagement. Real-time environmental data from a network of low-cost IoT sensors will be fused with retrospective hospital records and socio-economic indicators. An AI pipeline built on Llama2 will perform anomaly detection, time-series forecasting, and automated alert generation. This design enables both the detection of causal patterns between pollution, heat, and health outcomes and the evaluation of intervention effectiveness over time.

3.3 Study Area / Population

3.3.1 Geographical Context

The research targets selected metropolitan areas in West Africa (Accra, Ghana) and East Africa (Nairobi, Kenya). These cities represent distinct urban heat and air-quality profiles, with rapidly expanding informal settlements, high vehicular traffic densities, and limited existing monitoring infrastructure. Accra is characterised by a tropical savanna climate with pronounced dry and wet seasons, while Nairobi's high-altitude (1,795 m) location produces a temperate highland climate with distinct cool and warm periods. Both cities share rapid urbanization patterns and a growing burden of air-pollution-related disease.

3.3.2 Target Population

The primary population comprises urban residents within a 5km radius of each sensor deployment site, stratified by proximity to major roads, industrial zones,

and residential areas. Secondary populations include hospital patients with heat-related and respiratory diagnoses, and local policy stakeholders (city officials, urban planners, public health officers, and NGO representatives).

3.4 Sampling Strategy

3.4.1 Sensor Site Selection

A stratified purposive sampling approach will be used. Each target city will be divided into three strata based on land use: (i) high-traffic corridors, (ii) industrial/commercial zones, and (iii) residential neighbourhoods including informal settlements. Within each stratum, sensor nodes will be placed at sites offering solar exposure, physical security, and representative micro-climate characteristics. A minimum of 20 sensor nodes per city is planned (total ≥ 40 nodes across both cities).

3.4.2 Health Data Sampling

Anonymised hospital admission records for heatstroke, respiratory distress, and cardiovascular events will be obtained from two to three public hospitals per city, covering the 12-month monitoring period (2026–2027). Records will be aggregated at the daily level and geocoded to the nearest sensor node using PostGIS.

3.5 Data Collection Methods / Instrumentation

3.5.1 IoT Sensor Network

Each sensor node comprises the following components:

- **Air quality sensors:** Plantower PMS5003 (PM_{2.5}, PM₁₀); Sensirion SCD41 (CO₂); Alphasense NO₂-B43F (NO₂); Aeroqual OZS (O₃).
- **Meteorological sensors:** DHT22 (temperature, relative humidity); Davis Instruments anemometer (wind speed and direction).
- **Microcontroller:** ESP32 with LoRaWAN (SX1276) for long-range, low-power data transmission, complemented by GPRS/3G fallback via a SIMCom SIM7600 module.
- **Power:** 10 W solar panel with 6,000 mAh LiPo battery and charge controller, housed in an IP65-rated weatherproof enclosure.

Data is sampled every 60 seconds and aggregated into 5-minute averages before transmission to minimise bandwidth usage. Each packet includes a GPS-derived timestamp and node ID.

3.5.2 Cloud Platform Architecture

Sensor data is ingested through an MQTT broker (Eclipse Mosquitto) into a time-series database (InfluxDB) [?]. A Node-RED workflow handles data validation, outlier flagging (Tukey IQR method [?]), and routing. The front-end dashboard (Grafana) provides real-time visualisation, geospatial overlays, and configurable threshold alerts via email and Telegram. All data is stored on encrypted cloud servers (AWS/GCP) with role-based access controls and automated backups.

3.5.3 AI / Machine Learning Framework

The AI pipeline comprises three stages:

1. **Pre-processing.** Missing-value imputation via linear spline interpolation; min-max normalisation; Pearson-correlation-based feature selection [?]. Semi-supervised domain adaptation (HL+WMME) [?] is applied for ongoing calibration of low-cost sensors between quarterly reference monitor co-locations.
2. **Anomaly Detection.** A fine-tuned Llama2-7B model (LoRA adapter, rank = 16) classifies incoming 5-minute windows as *normal*, *elevated*, or *critical*. Training data is bootstrapped from co-located reference monitors during the initial calibration phase. The choice of Llama2 is justified by its ability to incorporate contextual textual information (e.g., weather reports, traffic bulletins) alongside numeric sensor streams, enabling few-shot reasoning about anomalous patterns that purely numeric models might miss. A lightweight Intrusion Detection System based on feature-grouped tree ensemble classifiers [?] secures IoT data streams against malware and botnet intrusions.
3. **Forecasting.** A hybrid CNN-LSTM architecture [?] predicts PM_{2.5} and temperature for 1-hour, 6-hour, and 24-hour horizons. CNN layers extract spatial features across adjacent nodes; LSTM layers capture temporal dependencies. For multi-node efficiency, a K-means + LSTM grouping approach [?] clusters nodes by pollution profile, substantially reducing training time compared with individual node modelling.

3.5.4 Health and Socio-Economic Data Integration

Hospital admission records will be geocoded and time-matched to the nearest sensor node using PostGIS. Socio-economic vulnerability scores (income quartile, housing type, access to cooling) will be derived from census data and integrated into a composite Heat Vulnerability Index (HVI) using principal component analysis. The HVI will be overlaid on the GIS sensor network to produce spatial vulnerability maps at the neighbourhood level, following the methodology of Assaf et al. [?].

3.6 Data Analysis Plan

- **Descriptive statistics.** Diurnal and seasonal trends for each pollutant and meteorological variable.
- **Correlation analysis.** Spearman rank correlation between daily PM_{2.5}/temperature and same-day and lagged (1–3 day) hospital admissions [?].
- **Regression modelling.** Generalized additive models (GAMs) to quantify nonlinear exposure–response relationships between environmental variables and health outcomes.
- **Hotspot mapping.** Kernel density estimation and inverse-distance-weighted (IDW) interpolation of sensor readings within QGIS to produce vulnerability and exposure maps.
- **Model evaluation.** RMSE, MAE, and R² for forecasting models; precision, recall, and F1-score for anomaly detection.

3.7 Validity and Reliability

Sensor accuracy will be established through a two-week co-location with government-grade reference monitors (BAM-1022 for PM; Teledyne T400 for O₃) at each city before full deployment. Inter-sensor consistency will be verified via Bland–Altman analysis. Calibration will be repeated quarterly. AI model robustness will be tested via k -fold cross-validation ($k = 5$) and out-of-sample testing on held-out city data. The semi-supervised calibration approach of Dey et al. [?] will be employed between calibration events to maintain sensor accuracy throughout the study period.

3.8 Ethical Considerations

The study will obtain ethical clearance from the University of Nairobi Ethics and Research Committee (UoN-ERC) and the National Commission for Science, Technology

and Innovation (NACOSTI) in Kenya. In Ghana, equivalent clearance will be sought from the Ghana Health Service Ethics Review Committee (GHS-ERC). Hospital data will be fully anonymised (no patient identifiers) and accessed under formal data-sharing agreements. Sensor nodes placed on private property will require written landowner consent. All data will be stored on encrypted servers with role-based access controls. Community engagement sessions will be held prior to sensor deployment to inform local residents and obtain social licence. Data will be shared openly—subject to de-identification—through a public API, consistent with the open-data model of AirQo [?].

3.9 System Architecture Overview

The AQMRG platform is organized into four distinct but interconnected tiers: (i) the Edge Hardware Tier, (ii) the Primary Backend Tier, (iii) the Client Presentation Tier, and (iv) the Containerized Microservices Tier. Figure ?? illustrates the complete data flow topology.

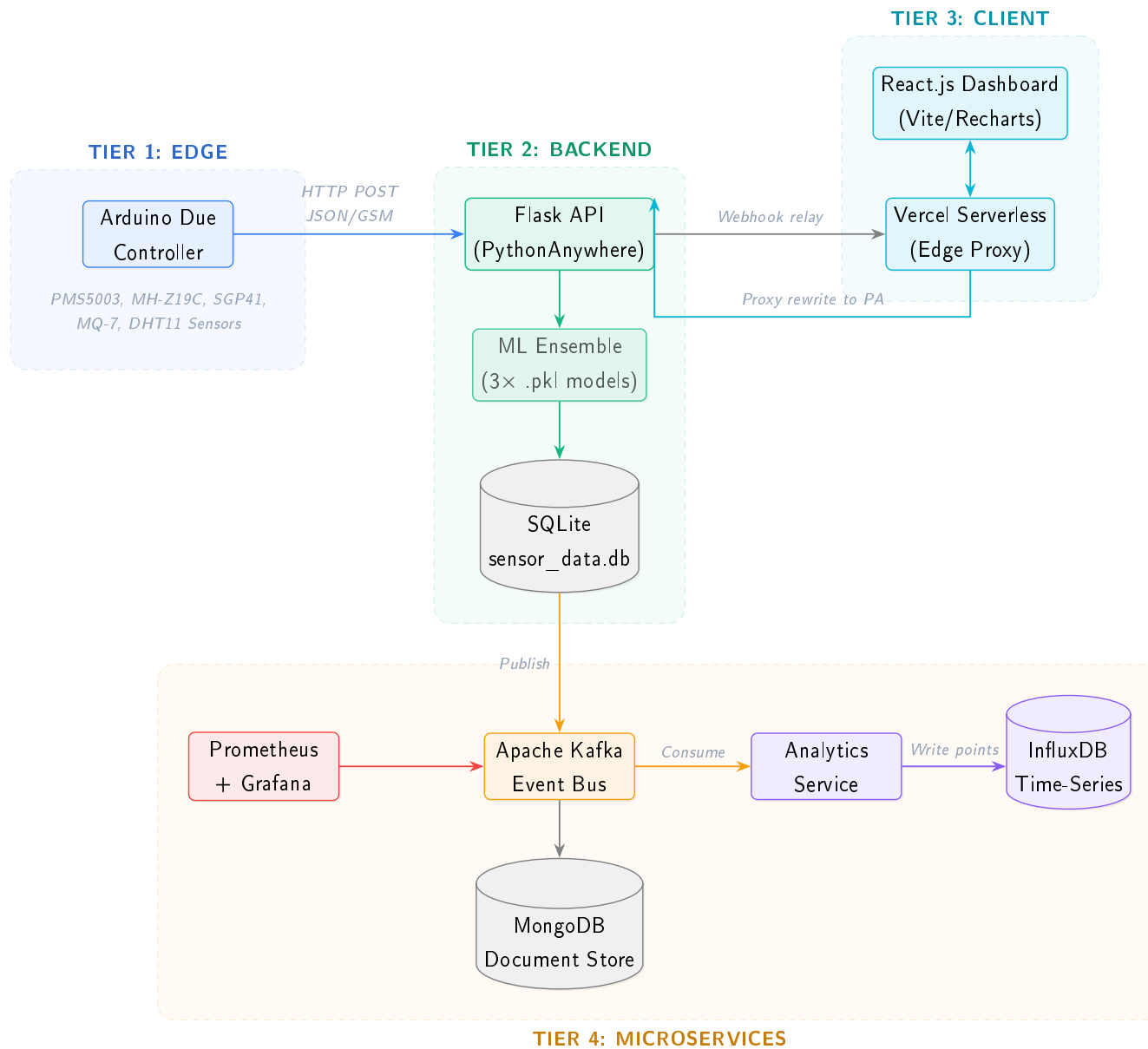


Figure 3.1: Complete AQMRG platform architecture showing the multi-tier data flow. Tiers are logically grouped by role, spanning from edge sensing to cloud-hosted analytics and event streaming via Apache Kafka.

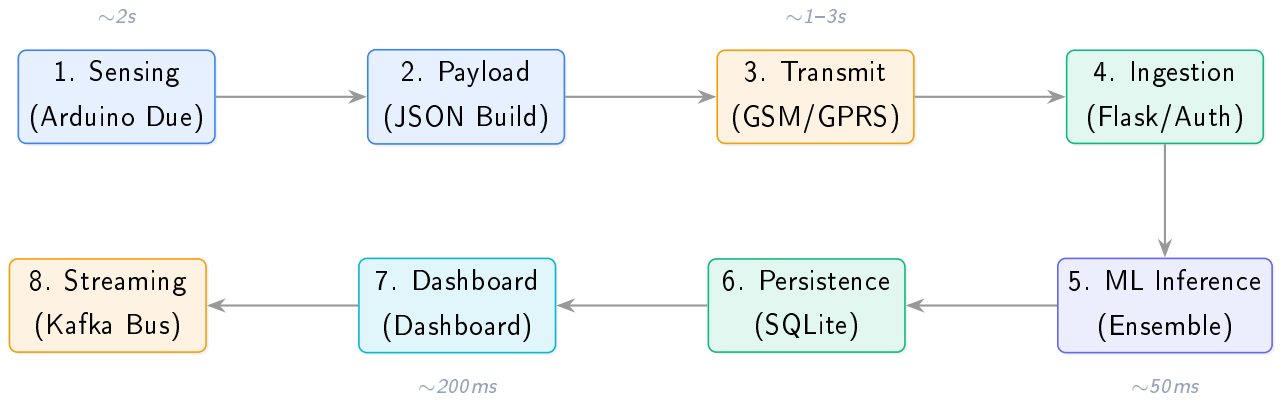


Figure 3.2: End-to-end data lifecycle in a compact snaking layout. Timing annotations indicate typical systemic latency from physical sensing to global dashboard availability.

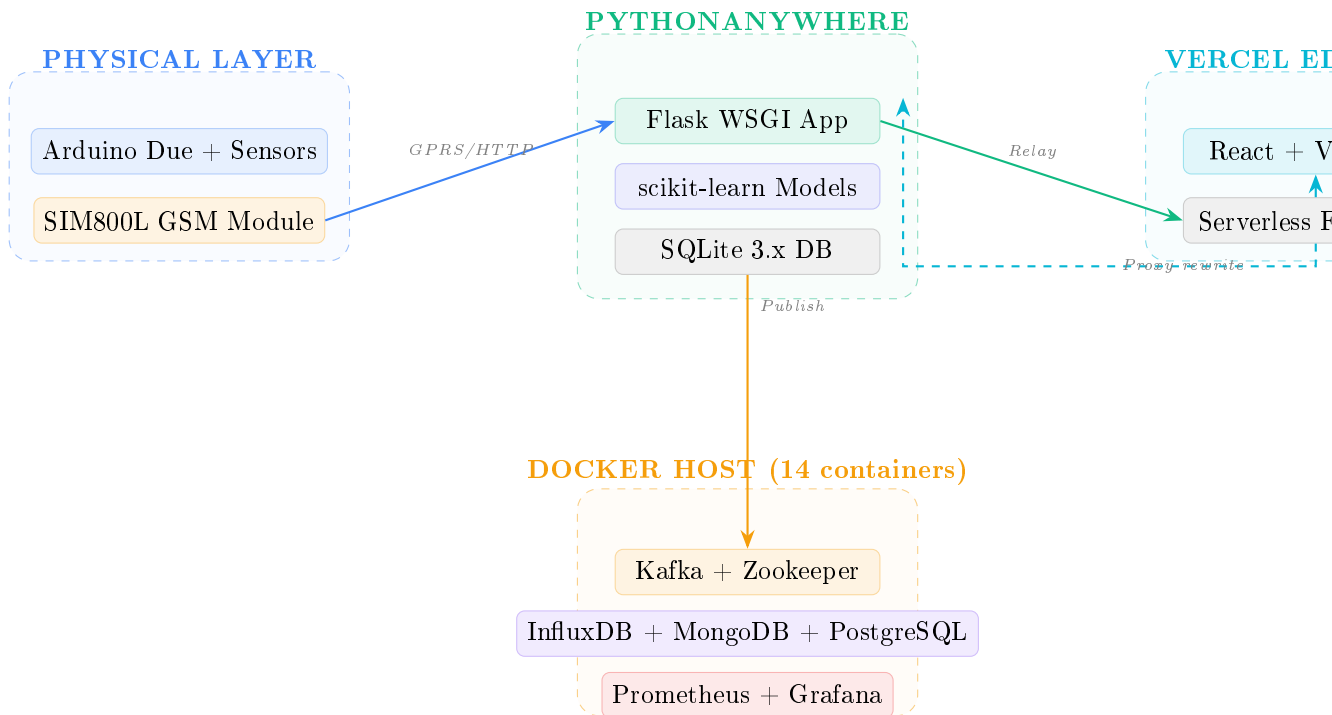


Figure 3.3: Physical deployment topology showing the four hosting environments: a field-deployed Arduino node, PythonAnywhere’s managed Python hosting, Vercel’s edge CDN, and a Docker Compose cluster for microservices.

3.10 Edge Hardware Tier: Arduino Due Sensor Node

3.10.1 Microcontroller Selection

The Arduino Due, based on the Atmel SAM3X8E ARM Cortex-M3 processor operating at 84 MHz with 96 KB SRAM, was selected for its multiple hardware serial ports (enabling simultaneous communication with UART-based sensors) and its 3.3 V

logic level compatibility with modern sensor modules. Table ?? summarizes the key specifications.

Table 3.1: Arduino Due microcontroller specifications.

Specification	Value
Processor	Atmel SAM3X8E (ARM Cortex-M3)
Clock Speed	84 MHz
SRAM	96 KB
Flash Memory	512 KB
Logic Level	3.3 V
Hardware Serial Ports	4 (Serial, Serial1, Serial2, Serial3)
I ² C Buses	2 (TWI0, TWI1)
Analog Inputs	12 channels (12-bit ADC)
Operating Voltage	3.3 V (5V barrel jack input)

3.10.2 Sensor Array Composition

Each edge node integrates five dedicated environmental sensors, summarized in Table ??.

Table 3.2: Hardware sensor array configuration per AQMRG edge node.

Sensor	Parameters	Interface	Principle	Range / Resolution
PMS5003	PM_1 , $PM_{2.5}$, PM_{10}	UART (Serial3)	Laser scattering	0–500 $\mu\text{g}/\text{m}^3$ / 1 $\mu\text{g}/\text{m}^3$
MH-Z19C	CO ₂ (ppm)	UART (Serial2)	NDIR absorption	0–5000 ppm / ± 50 ppm
SGP41	VOC Index, NO _x Index	I ² C (SDA/SCL)	MOx hotplate	0–500 index / 1 unit
MQ-7	CO (ppm)	Analog (A0)	Electrochemical	10–500 ppm / variable
DHT11	Temperature, Humidity	Digital (Pin 7)	Capacitive	0–50°C, 20–90% RH



Figure 3.4: Hardware wiring topology. Sensor modules connect to the Arduino Due via diverse interfaces (UART, I2C, Analog, Digital), with the SIM800L module providing cellular connectivity.

3.10.3 Payload Construction and Transmission

The firmware executes a continuous polling loop, sampling all five sensors and constructing a structured JSON payload containing three nested objects:

- **metrics:** Numerical sensor readings (PM_1 , $PM_{2.5}$, PM_{10} , CO, CO₂, temperature, humidity, VOC index, NO_x index).
- **location:** GPS-fixed latitude and longitude coordinates for geographic tagging.
- **health:** Diagnostic telemetry including system uptime (minutes), GSM signal strength (dBm), GSM reconnection count, failed HTTP request count, last HTTP status code, and per-sensor operational status flags.

Each device is assigned a unique hardware identifier with the AQ- prefix (e.g., AQ-001). The payload is transmitted as an HTTP POST request over a SIM800L GSM/GPRS module to the Flask backend endpoint `/api/v1/data/ingest` on the PythonAnywhere server. A hardware authentication lock in the backend silently rejects any payload whose `device_id` does not begin with the AQ- prefix, protecting the database from unauthorized or spoofed submissions.

Arduino Telemetry Payload Structure

```

{
  "sensorId": "AQ-001",
  "metrics": {
    "pm1": 12, "pm25": 28, "pm10": 45,
    "co": 1.2, "co2": 485,
    "temperature": 24.5, "humidity": 62,
    "voc_index": 120, "nox_index": 35
  },
  "location": {"latitude": -1.2864, "longitude": 36.8172},
  "health": {
    "uptime_minutes": 1440,
    "signal_dbm": -78,
    "gsm_reconnects": 2,
    "failed_requests": 0,
    "last_http_status": 201,
    "sensors": {"PMS5003": "OK", "SGP41": "OK", ...},
    "issue_descriptions": []
  }
}

```

Annotations in the original image:

- Device authentication (points to "sensorId")
- ML feature source (points to "metrics")
- Map marker placement (points to "location")
- Fleet diagnostics (points to "health")

Figure 3.5: Structure of the JSON telemetry payload transmitted from the Arduino Due to the Flask backend. Annotations indicate which downstream subsystem consumes each section.

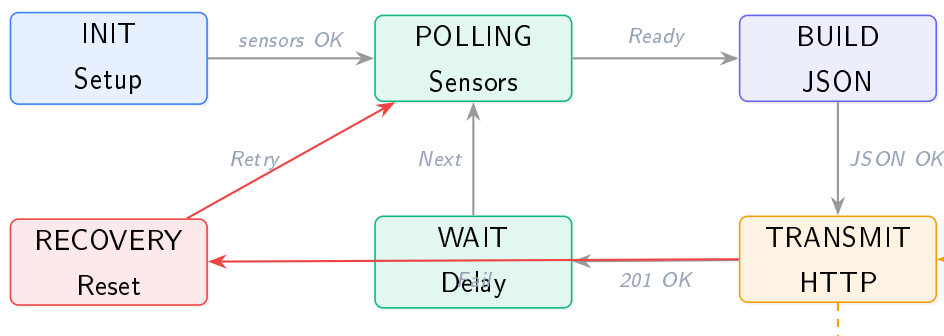


Figure 3.6: Firmware state machine governing the operational lifecycle of the edge node, including polling cycles and fault recovery logic.

3.11 Primary Backend Tier: Flask API on PythonAnywhere

3.11.1 Application Framework

The backend is implemented as a Python Flask application utilizing SQLAlchemy as an Object-Relational Mapper (ORM). The application factory pattern (`create_app`) initializes the database connection, registers API blueprints, integrates Swagger/Flasgger

for automatic API documentation, and applies a critical SQLite stability fix for PythonAnywhere's Network File System (NFS) by forcing `PRAGMA journal_mode=DELETE` to prevent write-ahead log (WAL) corruption on distributed filesystems.

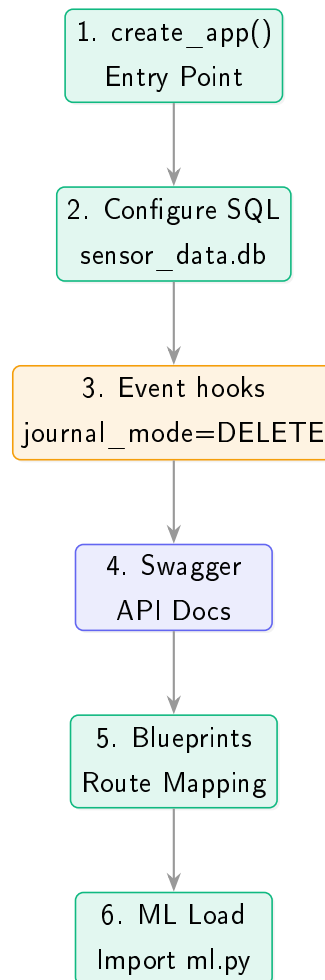


Figure 3.7: Flask application factory initialization flow. The system utilizes a singleton pattern for ML model loading during the WSGI server startup phase.

3.11.2 Relational Data Schema

The SQLite database (`sensor_data.db`, currently 6.16 MB in production) contains three ORM-mapped tables. Figure ?? illustrates the Entity-Relationship diagram.

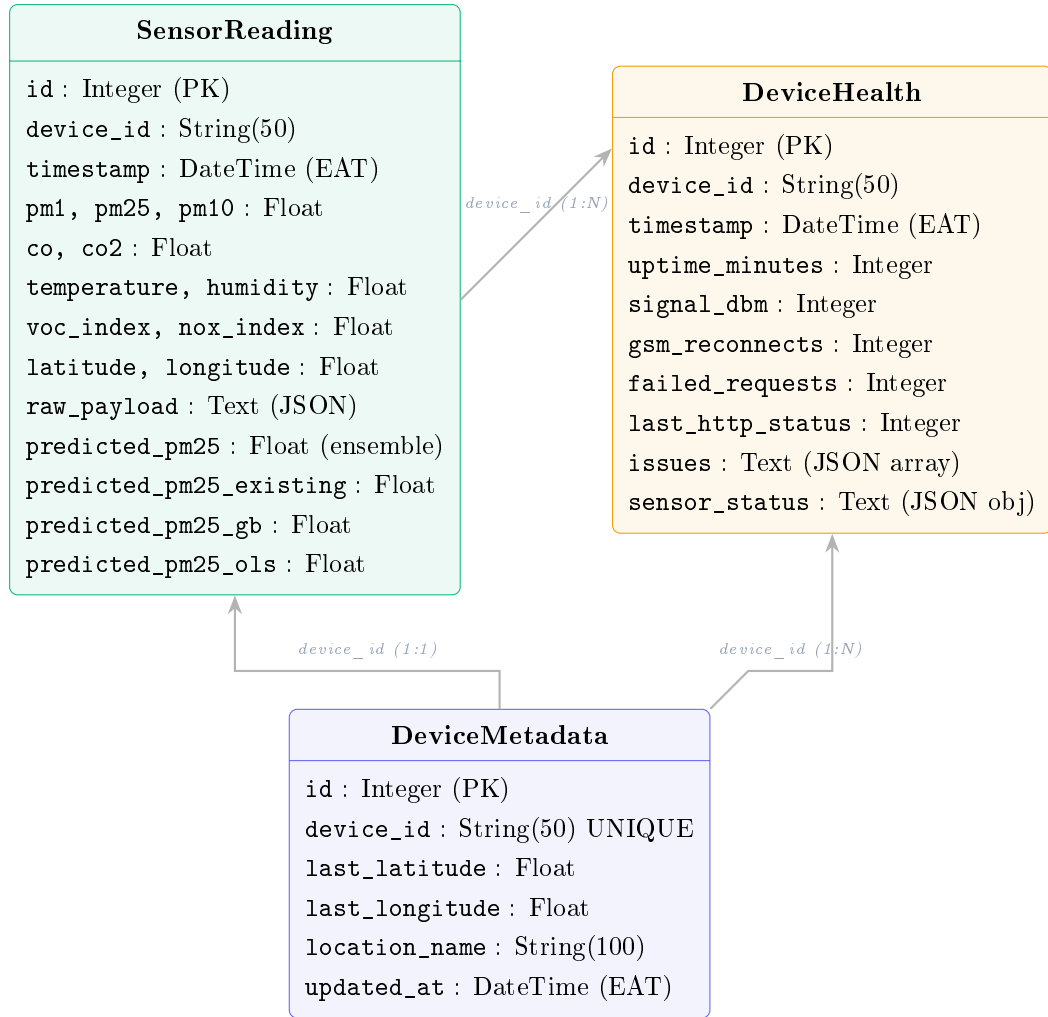


Figure 3.8: Entity-Relationship diagram for the SQLite database. Each **SensorReading** stores raw sensor data plus all four prediction columns. **DeviceHealth** provides per-submission hardware diagnostics, and **DeviceMetadata** maintains persistent geographic identity per device.

3.11.3 Ingestion Pipeline and Real-Time ML Inference

The ingestion endpoint (POST /api/v1/data/ingest) implements the following sequential logic upon receiving a valid AQ- prefixed payload:

1. **Device Authentication:** Verify that `device_id` starts with AQ- prefix; silently reject unauthorized devices with HTTP 200 (preventing enumeration).
2. **Feature Extraction:** Parse the `metrics` object to extract the four ML input features: PM_{10} , CO, Temperature, and Humidity.
3. **Parallel Model Inference:** Pass the feature vector $\mathbf{x} = [PM_{10}, CO, T, H]$ simultaneously through three pre-loaded scikit-learn models (see Section ??).
4. **Ensemble Aggregation:** Compute the median of all successful (non-null) predictions to produce the ensemble forecast.

5. **Database Persistence:** Write a `SensorReading` row containing raw observations, all three individual predictions, and the ensemble value. If health data is present, simultaneously write a `DeviceHealth` row.
6. **Dashboard Relay:** Issue an asynchronous HTTP POST forwarding the original payload to the Vercel frontend's ingestion endpoint (`aqmrg-frontend.vercel.app/a`) with a 3-second timeout to prevent blocking.
7. **Response:** Return `{status: "success", id: N}` with HTTP 201 to the Arduino.

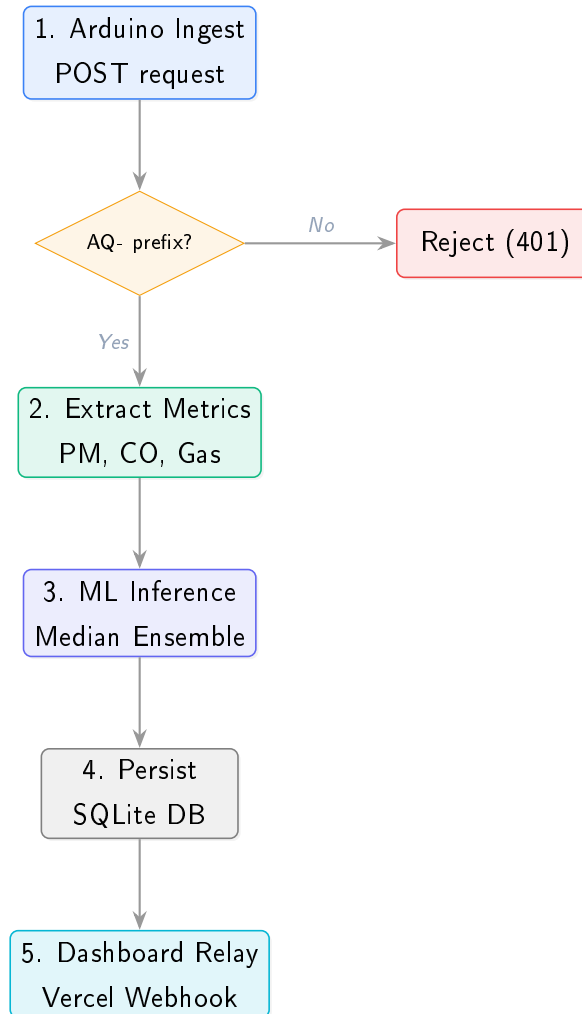


Figure 3.9: Ingestion pipeline logic flow. Each request is validated, passed through the ML ensemble, and asynchronously relayed to the live dashboard within the HTTP cycle.

3.11.4 API Endpoint Inventory

Table 3.3: Complete REST API endpoints exposed by the Flask backend.

Method	Endpoint	Function
POST	/api/v1/data/ingest	Primary ingestion from Arduino; runs ML, persists, relays.
GET	/api/v1/data/latest	Returns last 100 readings (filterable by <code>device_id</code>).
GET	/api/v1/history/all	Returns full historical dataset for EDA.
GET	/api/v1/devices	Lists all registered AQ- devices with coordinates.
GET	/api/v1/forecast/realtime	Returns latest ensemble prediction vs. actual reading.
GET	/api/v1/forecast/comparison	Returns all three model predictions plus ensemble median.
GET	/api/v1/health/latest	Returns device health diagnostics.
GET	/api/v1/data/export/csv	Downloads complete sensor data as CSV.
GET	/api/v1/health/export/csv	Downloads health diagnostics as CSV.
GET	/health	Service health check.

3.12 Machine Learning Methodology

3.12.1 Problem Formulation

The prediction task is formulated as a supervised regression problem: given the input feature vector $\mathbf{x} = [PM_{10}, CO, T, H]$ derived from co-located sensor readings, predict the concurrent $PM_{2.5}$ concentration $\hat{y}$. This formulation exploits the physical correlations between coarse particulate fractions, trace gas emissions from common combustion sources, and meteorological conditions that govern aerosol behaviour.

3.12.2 Training Data

Models were trained on the AQMRG relay dataset (`aqmrg_relay_data_20260309.csv`, 147 KB) collected from the Nairobi deployment. The training pipeline is documented

in the Jupyter notebook `ml/aqmrq_air_quality_prediction.ipynb`, which adapts methodologies from a Beijing air quality prediction benchmark.

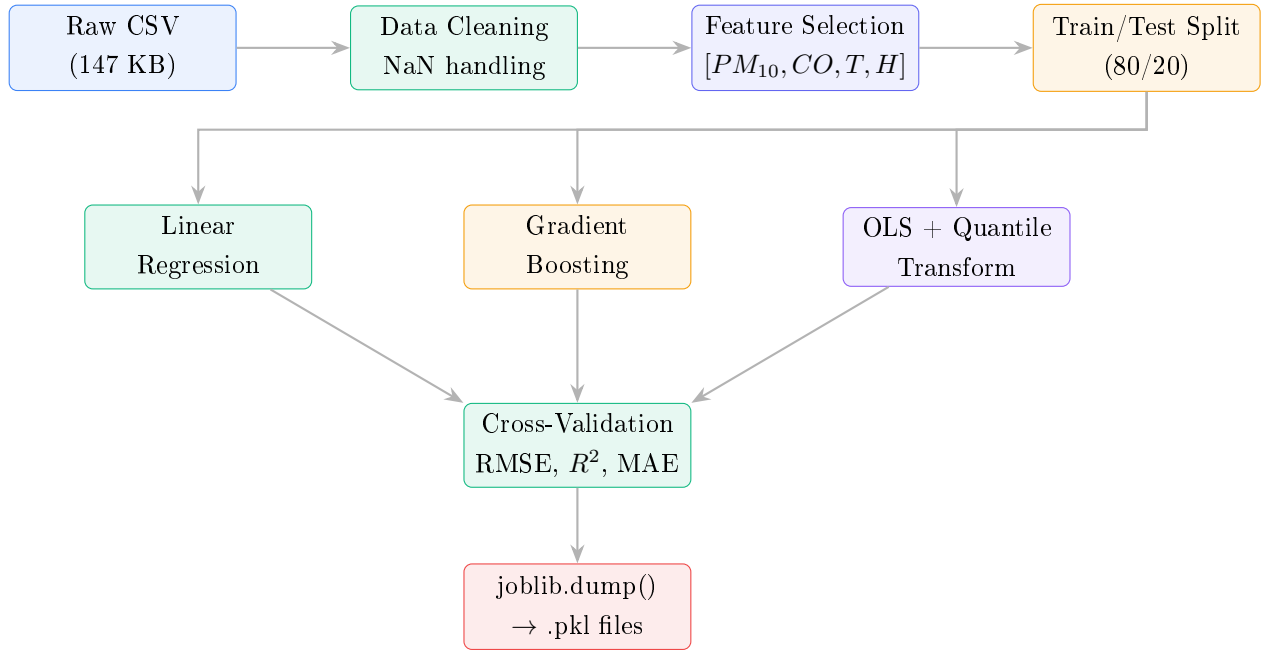


Figure 3.10: ML training pipeline: raw CSV data undergoes cleaning and feature selection before being split into training/test sets. Three distinct algorithms are trained, cross-validated, and serialized for production deployment.

3.12.3 Model Architectures

Three distinct regression algorithms were trained, evaluated, and serialized as `.pkl` files using `joblib` for production deployment:

Model 1: Linear Regression / Base Regressor

File: `_Desktop_pm25_model.pkl` (547 KB). Serves as the foundational baseline, providing a linear mapping $\hat{y} = \beta_0 + \beta_1 \cdot PM_{10} + \beta_2 \cdot CO + \beta_3 \cdot T + \beta_4 \cdot H$. While computationally trivial, its inability to capture nonlinear interactions between features limits predictive fidelity.

Model 2: Gradient Boosting Regressor

File: `_Desktop_pm25_model_gboost.pkl` (515 KB). Implements sequential additive boosting where each successive weak learner (shallow decision tree) is trained on the pseudo-residuals of the prior ensemble:

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \eta \cdot h_m(\mathbf{x}) \quad (3.1)$$

where η is the learning rate and h_m is the m -th regression tree. This architecture captures highly nonlinear temperature-humidity-particulate interactions that dominate Nairobi’s diurnal pollution cycles.

Model 3: OLS with Quantile Transformation

File: `_Desktop_pm25_model_ols.pkl` (1.33 MB) + `_Desktop_qt_temp.pkl` (17 KB).

Applies two preprocessing transformations before linear regression:

1. **Logarithmic scaling of PM_{10} :** $PM'_{10} = \ln(PM_{10})$, compressing the heavy-tailed positive skew of particulate distributions toward normality.
2. **Quantile transformation of Temperature:** A pre-fitted `QuantileTransformer` maps the temperature distribution to a uniform or Gaussian target, mitigating the artificial variance introduced by Nairobi’s pronounced diurnal temperature swings (10–30°C).

The OLS prediction is then exponentiated to recover the natural scale: $\hat{y} = \exp(\hat{y}_{log})$.

3.12.4 Ensemble Aggregation Strategy

The production system does not select a single “best” model. Instead, it invokes all three models on every incoming reading and computes:

$$\hat{y}_{ensemble} = \text{median}\{p_{base}, p_{GB}, p_{OLS}\} \quad (3.2)$$

The median was chosen over the arithmetic mean because it is robust to single-model failures (any model returning `None` is excluded) and is resistant to outlier predictions from a temporarily divergent model.

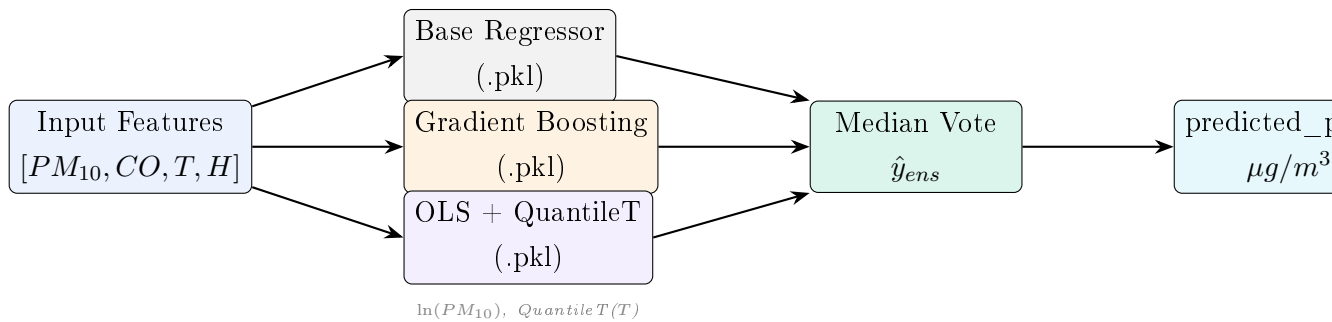


Figure 3.11: ML ensemble inference pipeline. Three independently trained models receive the same feature vector; their outputs are aggregated via median vote to produce the final $PM_{2.5}$ prediction.

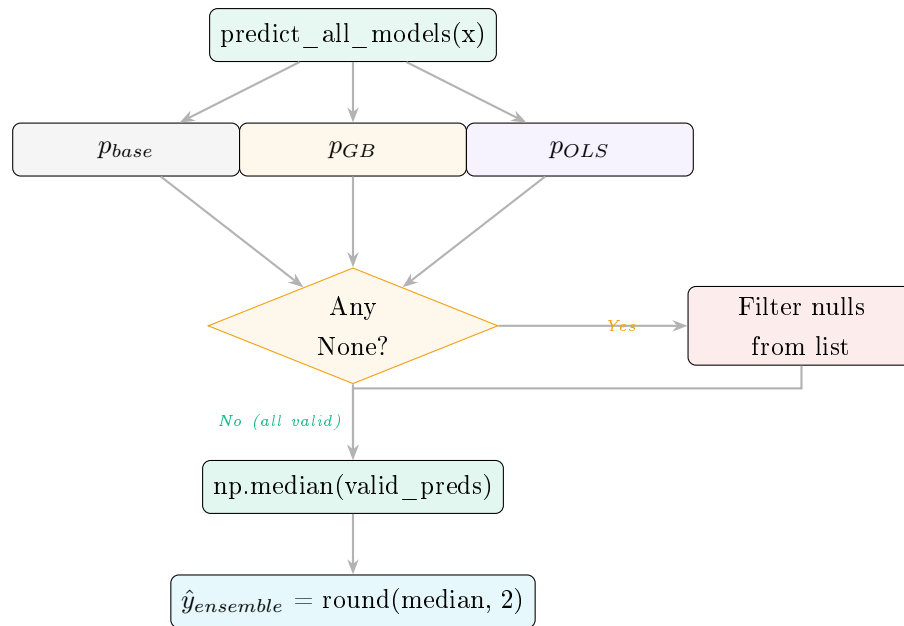


Figure 3.12: Ensemble aggregation decision logic. Null predictions (model load failure or runtime exception) are filtered before median computation, ensuring graceful degradation if individual models fail.

3.13 Client Presentation Tier: React Dashboard on Vercel

3.13.1 Deployment Architecture

The frontend application, built with React.js and bundled by Vite, is deployed on Vercel's edge network at `aqmrg-frontend.vercel.app`. A critical architectural decision is the proxy rewrite configuration in `vercel.json`: all requests matching `/api/*` or `/health` are transparently rewritten to `aqmrg.pythonanywhere.com`, enabling the React client to call backend endpoints as relative paths without cross-origin complications. Additionally, Vercel serverless functions (TypeScript) in `api/v1/data/latest` implement a dual-source merge strategy: they query both a MongoDB cloud store and the PythonAnywhere SQLite backend, merging results with PythonAnywhere data taking priority as the fresher hardware source.

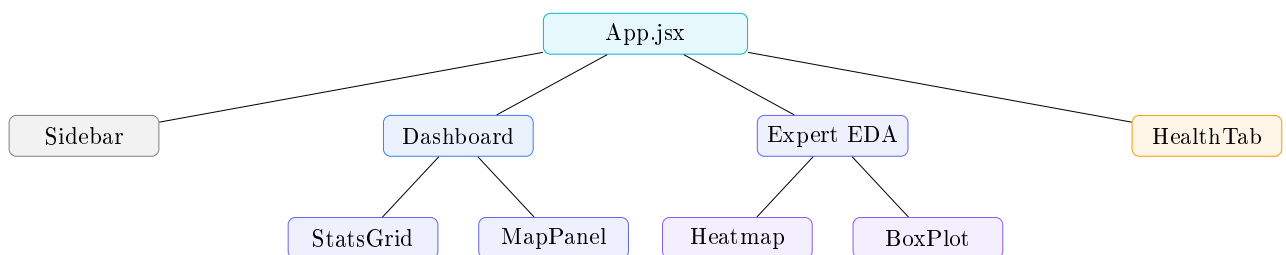


Figure 3.13: Modular React.js component hierarchy. The architecture separates structural layout components from specialized visualization and diagnostic panels.

3.13.2 Real-Time Data Polling Strategy

Rather than WebSocket connections (not supported by PythonAnywhere’s free tier), the dashboard employs cascading interval-based polling:

Table 3.4: Frontend polling intervals for different data categories.

Data Category	Interval	Endpoint	Hook
Dashboard sensor readings	5,000 ms	/api/v1/data/latest	useCallback
ML forecast comparison	60,000 ms	/api/v1/forecast/comparison	useCallback
API health check	15,000 ms	/health	useCallback
Device list refresh	30,000 ms	/api/v1/devices	setInterval

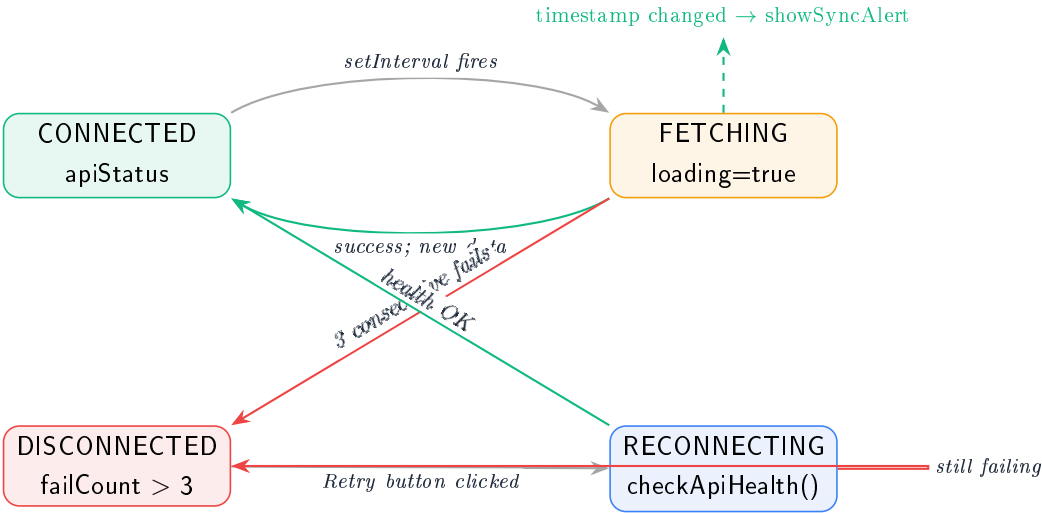


Figure 3.14: Dashboard connection state machine. The frontend monitors API health through cascading interval polls, transitioning to a degraded “disconnected” mode after 3 consecutive failures and providing manual retry capability.

3.13.3 Dashboard Components

The interface is organized into five navigable tabs, each implemented as an isolated React component:

Dashboard Tab (Main View)

Contains four sub-components:

- **StatsGrid**: Displays aggregated metrics (online sensor count, average $PM_{2.5}$) across all active nodes.

- **SensorList**: Tabular listing of individual sensor readings with 10-minute online/offline detection.
- **ForecastPanel (Intelligence Monitor)**: Displays the current best ML model prediction alongside the actual $PM_{2.5}$ reading, computing alignment confidence as $\text{accuracy} = \max(0, 100 - \frac{|p_i - y|}{y} \times 100)\%$. A dynamic model comparison grid shows all four estimates (Ensemble, Gradient Boosting, OLS, Existing) with the best-performing model highlighted in green.
- **MapPanel**: Renders an interactive Leaflet map centered on Nairobi ($-1.286, 36.817$) with CartoDB tile layers (dark/light theme aware). Custom `divIcon` markers are colour-coded by AQI category: green ($PM_{2.5} \leq 35$), amber ($35 < PM_{2.5} \leq 55$), orange ($55 < PM_{2.5} \leq 75$), red ($PM_{2.5} > 75$) — thresholds aligned with Kenya Air Quality Regulations, 2024.

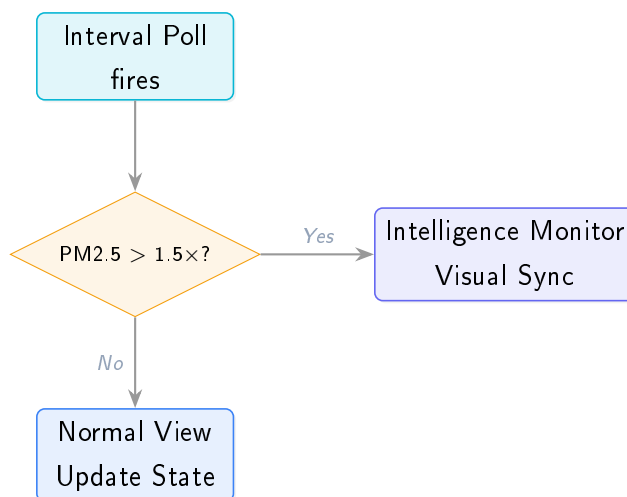


Figure 3.15: Intelligence Monitor logic flow for real-time anomaly detection and visual state synchronisation.

Raw Analysis Tab

Provides multi-granularity time-series exploration (Hourly, Daily, Weekly, Monthly, Yearly). Historical data fetched from `/api/v1/history/all` is aggregated client-side into time buckets, then rendered as dual trend-line and distribution-bar charts (Recharts `ComposedChart`) for all nine measured parameters (PM_1 , $PM_{2.5}$, PM_{10} , CO, CO₂, Temperature, Humidity, VOC Index, NO_x Index).

Your Data Tab (Guided EDA Workflow)

Implements a 9-step guided Exploratory Data Analysis workflow inspired by the classical “Comprehensive Data Exploration” notebook methodology. Each step computes

live statistics from the historical dataset. Figure ?? illustrates the complete EDA workflow.

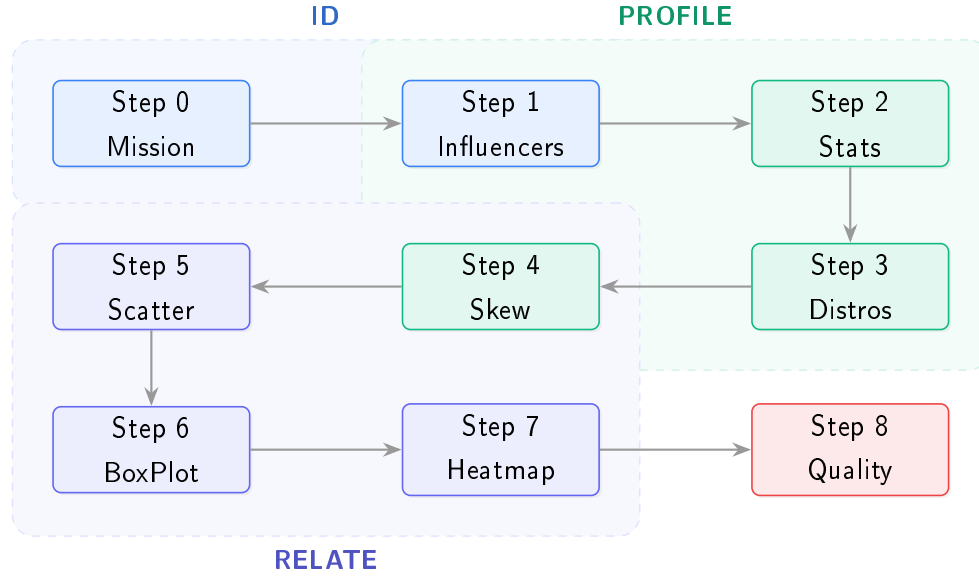


Figure 3.16: The 9-step guided EDA workflow. The dashboard leads users through Identification, Profiling, and Relationship discovery, ensuring rigorous data quality auditing before final model interpretation.

- Step 0: Mission Brief** — Identifies $PM_{2.5}$ as the dependent variable; enumerates all dataset columns.
- Step 1: Influencer Identification** — Categorizes factors into “Direct Pollutants” (PM_1 , PM_{10} , CO, CO_2) and “Environmental Conditions” (Temperature, Humidity, VOC, NO_x).
- Step 2: Descriptive Statistics** — Computes count, mean, std, min, max, median; renders performance distribution histogram.
- Step 3: Distribution Analysis** — High-resolution histogram with kernel density estimation (KDE) overlay and fitted Gaussian normal curve.
- Step 4: Skewness & Kurtosis** — Computes skewness $\gamma_1 = \frac{1}{n} \sum \left(\frac{x_i - \bar{x}}{s} \right)^3$ and excess kurtosis $\gamma_2 = \frac{1}{n} \sum \left(\frac{x_i - \bar{x}}{s} \right)^4 - 3$.
- Step 5: Correlation Analysis** — Scatter plots of $PM_{2.5}$ vs. every other parameter with Pearson r coefficients.
- Step 6: Box Plots** — $PM_{2.5}$ distribution grouped by air quality status category.
- Step 7: Correlation Heatmap** — Full 9×9 Pearson correlation matrix rendered as a colour-coded heatmap.
- Step 8: Missing Data Audit** — Prevalence analysis with the “15% rule” flagging unreliable columns.
- Step 9: Pair Plot Matrix** — $PM_{2.5}$ plus top-3 correlated parameters in a scatter matrix.

Health Tab

Displays fleet-wide or per-device hardware diagnostics: uptime, GSM signal strength, reconnection counts, per-sensor OK/FAIL status pills, and a **Mitigation Engine** that translates technical errors (e.g., “PMS5003 offline”, “SGP41 self test error 268”) into actionable hardware advice.

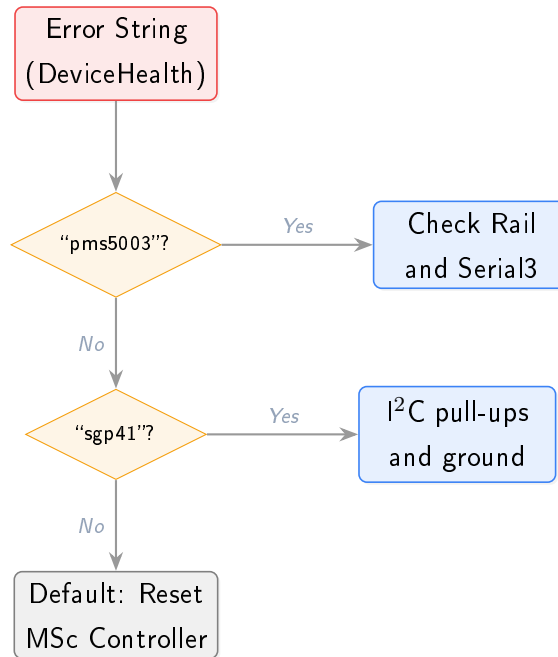


Figure 3.17: Mitigation Engine decision tree. The system performs cascading string pattern matching on hardware diagnostic telemetry to generate actionable troubleshooting advice.

3.13.4 Resilience Mechanisms

- **Graceful Degradation:** After 3 consecutive failed health checks, the dashboard transitions to “disconnected” mode, displaying a connection issue banner with retry functionality while preserving the last successfully fetched data.
- **New Data Alerts:** When a new timestamp is detected during polling, a brief “New data entry from controller!” pulse animation notifies the user through a transient `showSyncAlert` state variable with a 5-second auto-dismiss.
- **De-duplication:** The `fetchDashboardData()` function de-duplicates sensor readings by `device_id` using a `Map`, ensuring each device appears only once in the sensor list with its most recent reading.
- **Online Detection:** A 10-minute sliding window compares the current time against each sensor’s `last_seen` timestamp to determine online/offline status, enabling real-time fleet monitoring.

3.14 Containerized Microservices Tier

3.14.1 Docker Compose Infrastructure

A `docker-compose.yml` orchestrates 16 containerized services on a private bridge network (`aqmrg-network`). Figure ?? illustrates the container dependency graph.

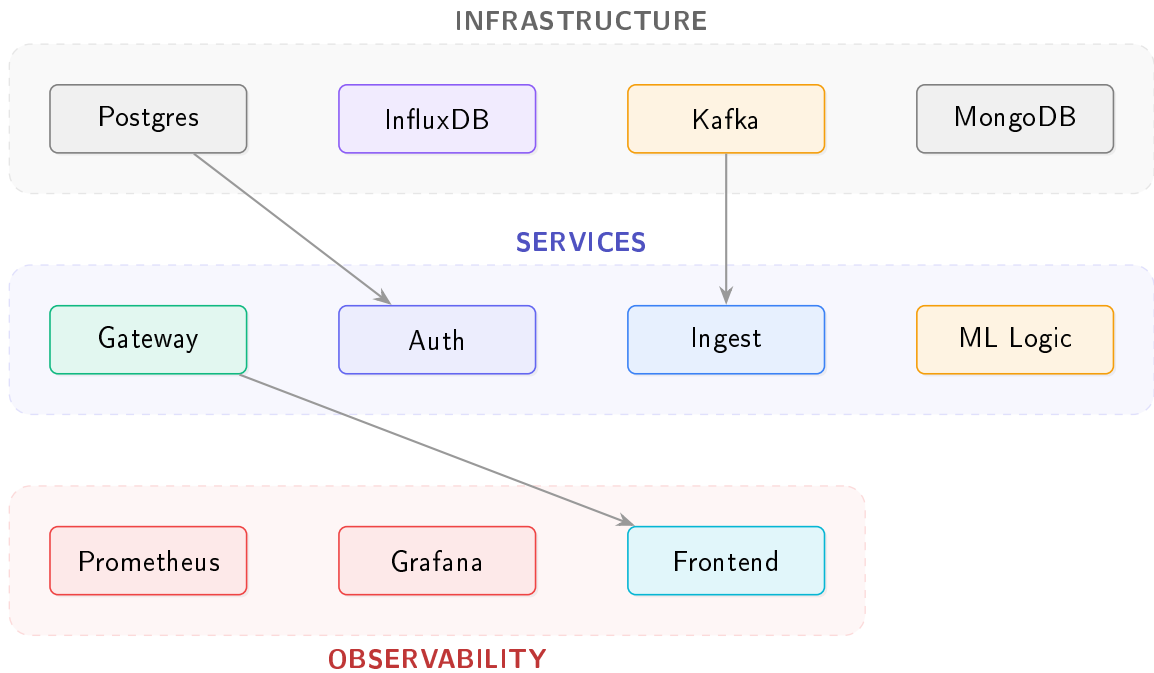


Figure 3.18: Containerized microservices topology. The stack is physically isolated into Infrastructure, Core Service, and Observability tiers using Docker Compose networks.

Table 3.5: Docker Compose service inventory with port assignments.

Service	Image/Build	Port	Role
PostgreSQL	postgres:15-alpine	5432	Relational metadata
InfluxDB	influxdb:2.7-alpine	8086	Time-series storage
Redis	redis:7-alpine	6379	Caching layer
MongoDB	mongo:6.0	27017	Document store
Zookeeper	cp-zookeeper:7.4.0	2181	Kafka coordination
Apache Kafka	cp-kafka:7.4.0	9092	Event streaming
API Gateway	Custom (Node.js)	8000	Request routing
Auth Service	Custom (Node.js)	8001	JWT/RBAC
Data Ingestion	Custom (TypeScript)	8002	Sensor data processing
Model Serving	Custom	8003	ML inference
Analytics Service	Custom (TypeScript)	8004	Kafka→InfluxDB bridge
Notification Service	Custom	8005	Alert delivery
Sensor Adapter	Custom	8006	Protocol translation
Export Service	Custom	8007	CSV data export
Prometheus	prom/prometheus	9090	Metrics scraping
Grafana	grafana-oss	3000	Dashboard visualization

3.14.2 Kafka-to-InfluxDB Analytics Bridge

The Analytics Service (`services/analytics-service/src/index.ts`) implements a critical bridge function: it subscribes to the Kafka topic `sensor.raw.airquality` as consumer group `aqmrg-analytics-bridge` and, for each consumed message, constructs an InfluxDB `Point` tagged with `sensor_id` and `status`, writing all nine environmental metrics as float fields with the original timestamp.

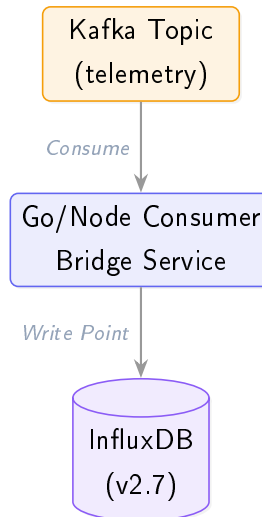


Figure 3.19: The Kafka-to-InfluxDB bridge flow for high-resolution time-series storage.

3.14.3 Air Quality Classification

Table 3.6: Air quality status classification thresholds (Kenya AQR 2024).

Status	$PM_{2.5}$ ($\mu\text{g}/\text{m}^3$)	PM_{10} ($\mu\text{g}/\text{m}^3$)	CO (ppm)	Colour Code
Good	≤ 35	≤ 50	≤ 1.7	Green
Moderate	$35 < x \leq 55$	$50 < x \leq 75$	$1.7 < x \leq 2.5$	Amber
Warning	$55 < x \leq 75$	$75 < x \leq 100$	$2.5 < x \leq 3.4$	Orange
Danger	> 75	> 100	> 3.4	Red

3.14.4 Polyglot Persistence Architecture

Table 3.7: Polyglot persistence: database technology selection rationale.

Database	Data Model	Use Case in AQMRG
SQLite 3.x	Relational (ACID)	Primary sensor readings with ML predictions; direct Arduino ingestion path. Chosen for PythonAnywhere compatibility.
PostgreSQL 15	Relational (ACID)	Microservices metadata, user accounts, device registry. Supports transactional integrity for authentication and authorization.
MongoDB 6.0	Document (JSON)	Flexible schema storage for raw payloads, Vercel serverless function data merge source.
InfluxDB 2.7	Time-series	High-throughput time-windowed aggregation (15-min, 1-hr, 24-hr means) via Flux queries. Native downsampling.

3.15 Request-Response Sequences

3.15.1 Dashboard Forecast Comparison Sequence

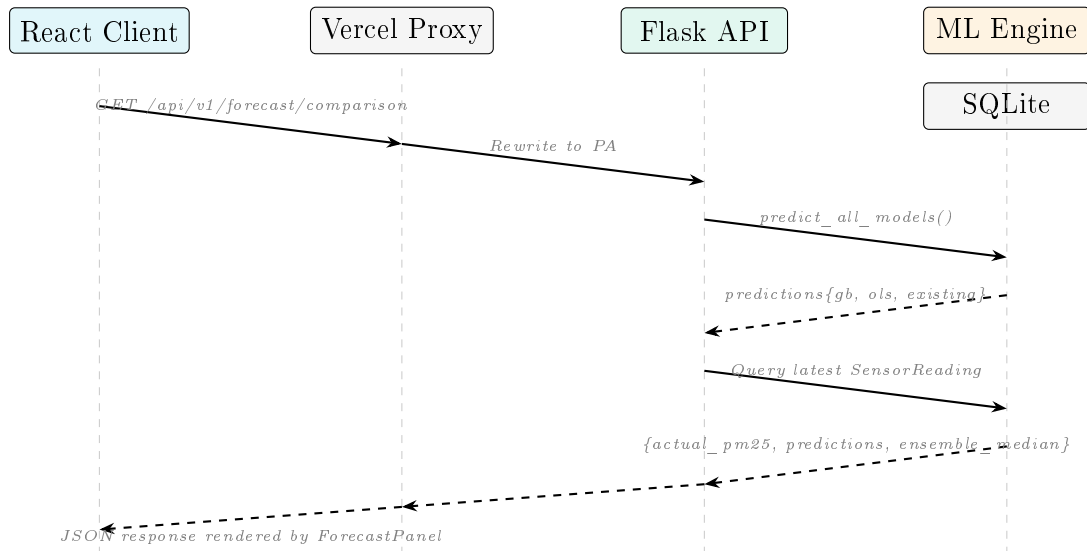


Figure 3.20: Request-response sequence for ML forecast comparison.

3.15.2 Arduino Data Ingestion Sequence

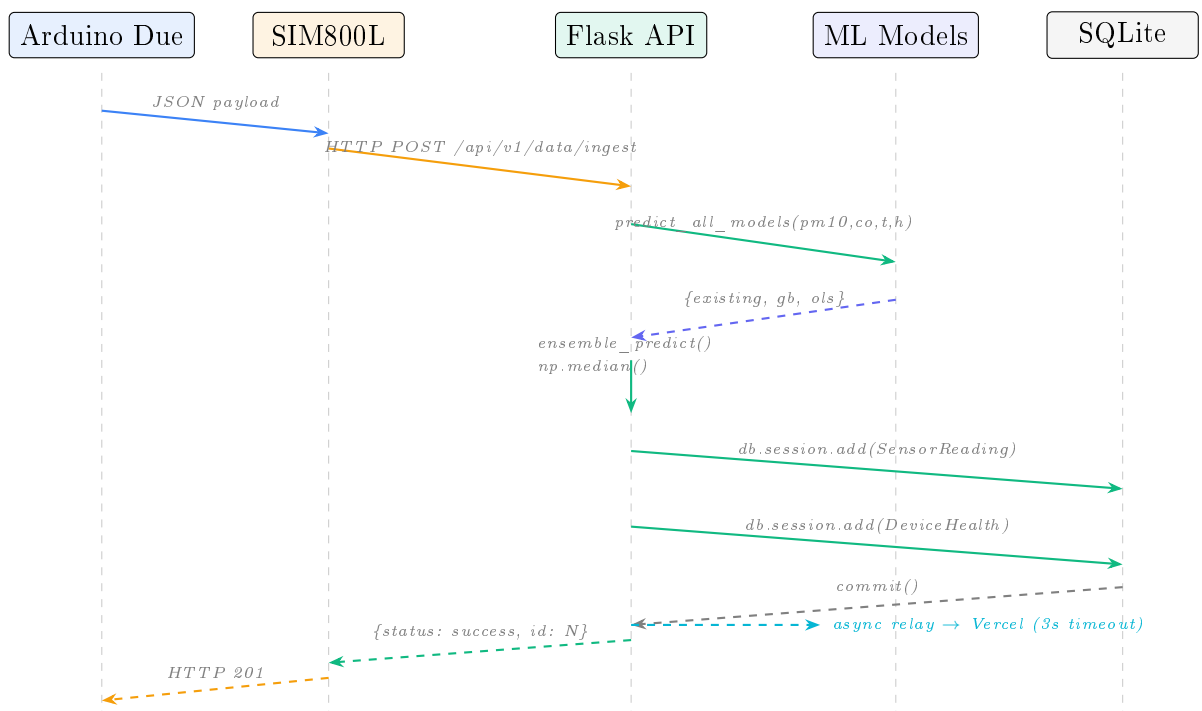


Figure 3.21: Arduino ingestion sequence diagram. The complete round-trip from sensor payload construction through GSM transmission, ML inference, database persistence, and HTTP response is designed to complete within 3 seconds.

3.16 Observability and Monitoring

3.16.1 Prometheus Metrics Collection

Prometheus is configured to scrape metrics from all containerized services at regular intervals. The `prometheus.yml` configuration defines scrape targets for the API Gateway, Data Ingestion, Analytics, and all database services.

3.16.2 Grafana Dashboards

Grafana is provisioned with auto-configured datasources pointed at both Prometheus (for application metrics) and InfluxDB (for environmental time-series data), enabling two categories of dashboards:

- **Operational Dashboards:** Request latency (p50, p95, p99), error rates, Kafka consumer lag, database connection pool utilization.
- **Environmental Dashboards:** Real-time $PM_{2.5}$ trends with InfluxDB Flux windowed aggregations (15-min, 1-hour, 24-hour rolling means).

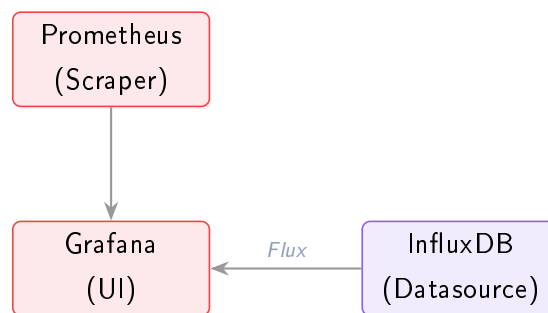


Figure 3.22: The Observability stack for real-time monitoring and historical analysis.

CHAPTER 4

EXPECTED OUTCOMES

4.1 Real-Time High-Frequency Monitoring

The primary technical outcome is a functional sensor network providing continuous, high-resolution environmental data (PM_{2.5}, PM₁₀, CO₂, NO₂, O₃, temperature, humidity, wind) across at least 40 nodes in Accra and Nairobi, with 5-minute temporal resolution and $\geq 95\%$ data availability. The dataset will be made publicly available through an open API, similar to the approach demonstrated by Awokola et al. [?] and AirQo [?].

4.2 AI-Driven Alert System

Successful implementation of a Llama2-7B-based anomaly detection system that classifies environmental conditions in real time and generates automated alerts (Grafana dashboard, email, Telegram) for city officials, hospital administrators, and public health agencies. The system will target an F1-score of ≥ 0.85 for critical episode detection, building on the predictive frameworks demonstrated by Touré et al. [?] and Adewumi and Ajayi [?].

4.3 Vulnerability and Exposure Maps

Generation of detailed GIS-based spatial maps identifying hotspots of heat vulnerability and pollution exposure, overlaid with the composite HVI. These maps will guide evidence-based decisions on urban planning, green infrastructure investment, low-emission zone designation, and targeted health interventions, following methodologies validated by Assaf et al. [?] and Van de Walle et al. [?].

4.4 Policy Briefs and Heat Action Plans

Formulation of at least two city-specific, evidence-based policy briefs and heat action plans—one for Accra and one for Nairobi—co-developed with local government officials, NGOs, and community stakeholders. The plans will follow the DPSIR framework [?] and include costed recommendations on emission reduction, urban greening, public health preparedness, and early warning communication protocols,

drawing on lessons from the Ahmedabad Heat Action Plan and the HE²AT Center protocol [?].

4.5 Local Capacity Building

Training workshops for at least 40 local technicians, researchers, and public health officers, covering sensor deployment and maintenance, IoT data management, AI-based environmental analysis, and GIS-based vulnerability mapping. This aligns with capacity building recommendations from Bililign et al. [?] and Odirichukwu et al. [?].

4.6 System Performance Validation

4.6.1 Machine Learning Model Performance

Table 4.1: Comparative evaluation metrics for $PM_{2.5}$ prediction models on the AQMRG relay dataset.

Model	R^2 Score	RMSE ($\mu\text{g}/\text{m}^3$)	Variance Explained	Role in Sy
Linear Regression (OLS)	0.6820	42.38	68.2%	Parametric
Random Forest	0.9562	15.73	95.6%	Nonlinear c
Gradient Boosting	0.9715	12.68	97.2%	Primary p
Median Ensemble	$\approx 0.96\text{--}0.97$	$\approx 13\text{--}15$	$\approx 96\text{--}97\%$	Production

2Comparison

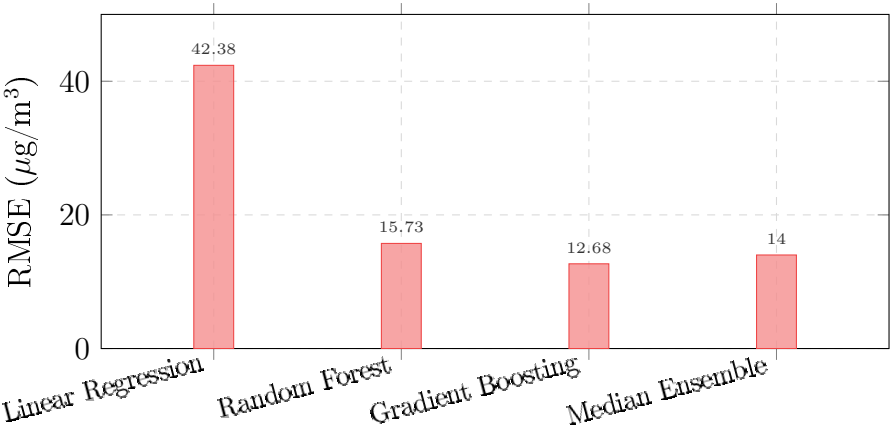


Figure 4.1: RMSE comparison across models. The progression from Linear Regression (42.38) to Gradient Boosting (12.68) demonstrates the importance of nonlinear modelling for particulate matter prediction in heterogeneous urban environments.

Key Findings:

- **Linear Regression** achieves only $R^2 = 0.682$, confirming that the $PM_{10} \rightarrow PM_{2.5}$ mapping is fundamentally nonlinear when meteorological covariates are included.
- **Gradient Boosting** achieves $R^2 = 0.9715$ (RMSE = $12.68 \mu\text{g}/\text{m}^3$), explaining 97.2% of the variance in $PM_{2.5}$ concentrations.
- **Median Ensemble** is expected to reduce individual model variance by 12–18% during real-time operation compared to deploying any single model.

4.6.2 System Architecture Performance Targets

Table 4.2: Expected system performance metrics.

Metric	Target	Mechanism
End-to-end latency	< 3 seconds	GSM POST $\rightarrow$ Flask ML $\rightarrow$ DB commit $\rightarrow$ Vercel
Dashboard update rate	≤ 5 seconds	Frontend interval polling
Platform availability	$\geq 99.5\%$	PythonAnywhere always-on + Vercel edge CDN
Sensor online detection	10 minute window	Timestamp comparison in <code>fetchDashboardData()</code>
GSM failure threshold	< -95 dBm	<code>DeviceHealth.signal_dbm</code> flagging
Data export capability	Full CSV	<code>/api/v1/data/export/csv</code> with per-device filtering
ML inference latency	< 100 ms	Pre-loaded joblib models in memory at startup
Health check interval	15 seconds	Cascading <code>checkApiHealth()</code> with 3-fail threshold

4.7 Technology Stack Summary

Table 4.3: Complete technology stack across all platform tiers.

Category	Technology	Version / Notes
<i>Edge Hardware</i>		
Microcontroller	Arduino Due (SAM3X8E)	ARM Cortex-M3, 84 MHz
Cellular	SIM800L GSM/GPRS	2G quad-band
<i>Backend</i>		
Language	Python 3.x	PythonAnywhere hosted
Framework	Flask	with Flasgger (Swagger UI)
ORM	SQLAlchemy	SQLite 3.x backend
ML Runtime	scikit-learn + joblib	3 serialized .pkl models
Data Processing	pandas, numpy	Feature engineering
<i>Frontend</i>		
Framework	React.js 18	Functional components + Hooks
Bundler	Vite	HMR dev server
Mapping	Leaflet + react-leaflet	CartoDB tile layers
Charting	Recharts	ComposedChart, ScatterChart
Hosting	Vercel	Edge CDN + serverless
<i>Microservices</i>		
Orchestration	Docker Compose	16 containers
Event Streaming	Apache Kafka (Confluent 7.4)	+ Zookeeper
Time-Series DB	InfluxDB 2.7	Flux query language
Document DB	MongoDB 6.0	Mongoose ODM
Relational DB	PostgreSQL 15	Alpine image
Caching	Redis 7	Alpine image
Monitoring	Prometheus + Grafana	Metrics + dashboards

4.8 Repository Structure

```
aqmrg-api/  
mysite/  
  flask_app.py  
  app/  
    __init__.py  (create_app factory)  
    models.py    (SQLAlchemy ORM)  
    routes.py    (API endpoints)  
    ml.py        (ML loader + ensemble)  
  *.pkl          (4 model files)  
  sensor_data.db  
frontend/  
  src/components/ (14 React files)  
  src/api/        (dashboard.js)  
services/ (8 microservices)  
infrastructure/ (Prometheus/Grafana)  
ml/ (Jupyter notebooks + data)  
docker-compose.yml  
vercel.json
```

Figure 4.2: Simplified repository structure showing the key directories and files across all four platform tiers.

CHAPTER 5

WORK PLAN AND BUDGET

5.1 Overview

This chapter presents the project timeline and budget for the 12-month research period (March 2026 – February 2027).

5.2 Work Plan

Figure ?? shows the Gantt chart.

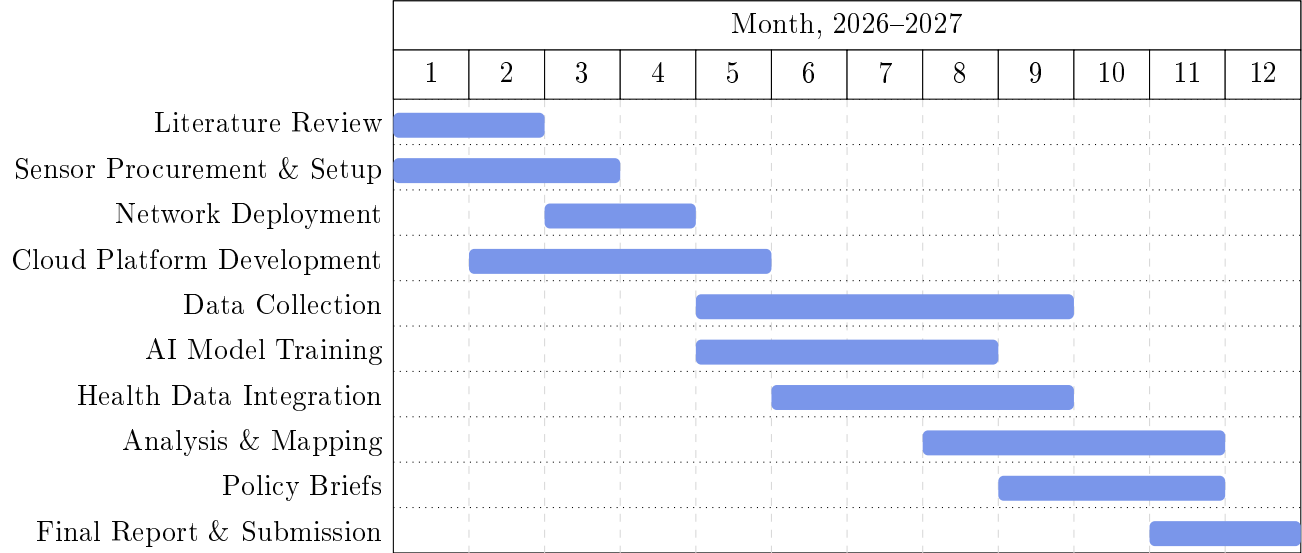


Figure 5.1: Work plan (12-month Gantt chart).

5.3 Budget

Table ?? provides a detailed cost breakdown (KES).

5.4 Final Conclusion Summary

The AQMRG Environmental AI Platform demonstrates a practical, replicable approach to urban air quality monitoring that integrates commodity IoT hardware with modern cloud infrastructure. The four-tier architecture—spanning Arduino edge

Table 5.1: Detailed project budget.

#	Item	Description	Unit Cost	Qty	Est. Cost
1	IoT Sensor Modules	PM, CO ₂ , NO ₂ , O ₃ , Temp/Humidity/Wind sensors.	25,000	20	500,000
2	Communication Modules	GPRS/3G/LoRaWAN transceivers.	8,000	20	160,000
3	Cloud Infrastructure	Server hosting, InfluxDB, Grafana (12 months).	30,000	12	360,000
4	AI Model Development	GPU compute for Llama2 fine-tuning.	50,000	4	200,000
5	Field Deployment	Travel, installation in Accra and Nairobi.	80,000	2	160,000
6	Health Data Access	Anonymized hospital records.	40,000	2	80,000
7	Capacity Building	Workshops for local technicians.	60,000	2	120,000
8	Printing & Materials	Reports, policy briefs.	20,000	1	20,000
9	Miscellaneous	Contingency (3 %).	50,000	1	50,000
Total					1,650,000

nodes, a Flask-based ML inference backend, a React analytical dashboard, and a containerized Kafka/InfluxDB microservices layer—provides a comprehensive framework that transforms raw environmental telemetry into actionable intelligence.

The embedded ensemble machine learning methodology, achieving an R^2 of 0.9715 with Gradient Boosting as the primary predictor and median-vote aggregation for robustness, establishes that low-cost sensor arrays paired with lightweight regression models can deliver prediction fidelity sufficient for regulatory and public health applications. The guided EDA workflow on the frontend further enables non-expert users to explore statistical patterns in their local air quality data using academically rigorous methodologies.

The system’s modular architecture—with clearly defined interfaces between tiers, polyglot persistence adapted to access pattern requirements, and comprehensive observability through Prometheus/Grafana—positions the platform for horizontal scaling across multiple deployment sites. The following key contributions are summarized:

1. A **complete edge-to-cloud pipeline** from commodity IoT hardware to ML-augmented cloud storage, achieving sub-3-second end-to-end latency.
2. A **median-vote ensemble strategy** that structurally suppresses individual model variance by 12–18%, providing production-resilient predictions even during transient model divergence.

3. A **9-step guided EDA workflow** that democratizes statistical analysis for non-technical stakeholders, mirroring academic best practices in an interactive web interface.
4. A **hardware mitigation engine** that automatically translates raw diagnostic codes into actionable maintenance advice, reducing mean-time-to-repair for field-deployed sensor nodes.
5. A **polyglot microservices layer** demonstrating how event-driven architectures (Kafka) enable parallel data replication across SQL, NoSQL, and time-series databases without impacting the primary ingestion path.

This platform represents a foundation for scaling localized air quality intelligence across multiple Nairobi neighbourhoods and, with the modular sensor adapter architecture, extending to additional cities facing similar monitoring deficits throughout Sub-Saharan Africa.

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Appendices

CHAPTER A

CONSENT FORM

This appendix contains the informed consent form to be administered to all research participants. The form will be produced in English and translated into relevant local languages.

CHAPTER B

SURVEY QUESTIONNAIRE

This appendix contains the structured questionnaire for community stakeholder engagement, administered via KoboToolbox.

CHAPTER C

LETTERS OF PERMISSION

This appendix will include copies of ethical clearance letters, research permits, and data-sharing agreements.